

An Advanced, Explainable AI Framework for Healthcare Fraud Detection with Agents and LLMs

M.Tech Project. 4th Semester (Review 2)

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Advanced Healthcare Fraud Detection System

1. Project Overview

1.1 Purpose

FraudGuard is an advanced healthcare fraud detection system designed to identify anomalous billing patterns in Medicare data using Multi-Agent AI, Deep Learning, and Explainable AI (XAI). The system analyzes provider behavior across multiple data sources to detect potential fraud, waste, and abuse.

1.2 Key Objectives

- Automated Fraud Detection: Identify fraudulent billing patterns using deep learning models
- Explainability: Provide clear, interpretable explanations for all predictions using SHAP
- Multi-Agent Collaboration: Employ specialized AI agents for investigation, analysis, and reporting
- Real-time Monitoring: Continuously scan provider database for emerging risks
- Scalability: Process large-scale Medicare datasets efficiently

1.3 Core Capabilities

- Deep neural network-based fraud risk scoring (0.0-1.0 scale)
- SHAP-based feature importance analysis
- Network analysis using provider similarity graphs
- Rule-based anomaly detection
- LLM-powered narrative report generation
- Real-time dashboard with search and filtering
- Session-based authentication system
- Model Context Protocol (MCP) integration

Abstract

Healthcare fraud, waste, and abuse (FWA) represents one of the most significant and persistent financial drains on global economies, with estimates suggesting that it accounts for 3-10% of all healthcare spending, translating to hundreds of billions of dollars annually. Beyond the financial impact, these illicit activities compromise patient safety, corrupt medical data, and erode public trust in healthcare systems.

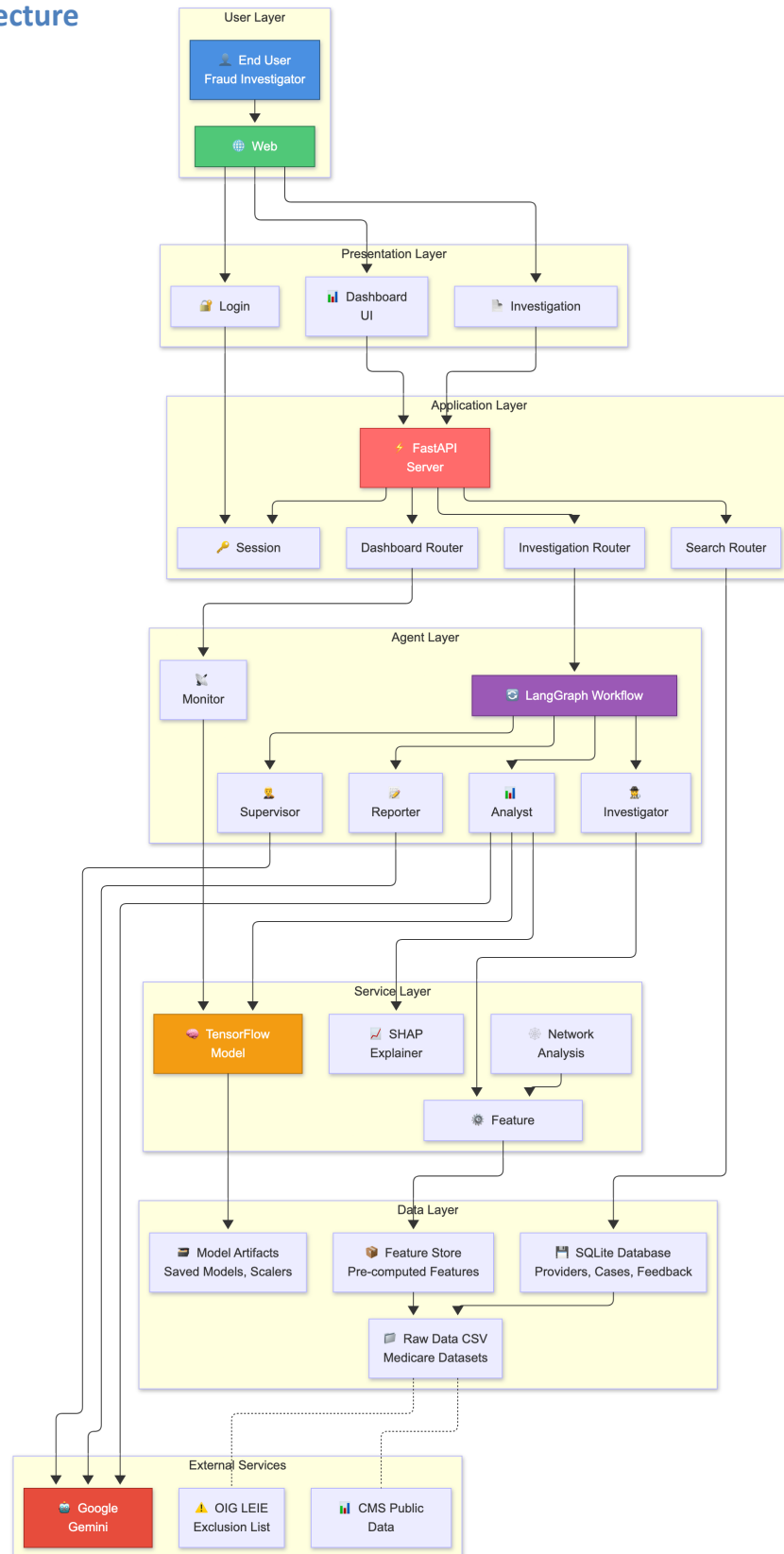
Traditionally, the fight against FWA has relied on post-payment, "pay-and-chase" models. These systems use static, rule-based engines and manual audits to identify suspicious claims after payments have already been disbursed. However, this approach is fundamentally reactive and struggles to keep pace with the increasing sophistication of fraudulent schemes. Modern fraud is often collusive, dynamic, and hidden within massive volumes of seemingly legitimate claims data.

This has led to a decisive shift in the research landscape towards proactive, data-driven detection methodologies powered by Artificial Intelligence (AI) and Machine Learning (ML). This survey critically assesses the state-of-the-art research in AI-driven healthcare fraud detection, focusing on the most recent and impactful publications from 2020 to 2025. We will explore the dominant research trends, summarize key papers within each trend, and provide a critical synthesis of the field's current trajectory, its challenges, and the gaps that contemporary research, including this project, aims to fill.

The scope of this survey is focused on provider-level fraud detection in large-scale claims databases (such as Medicare), as this represents the core of the academic and industrial research effort.

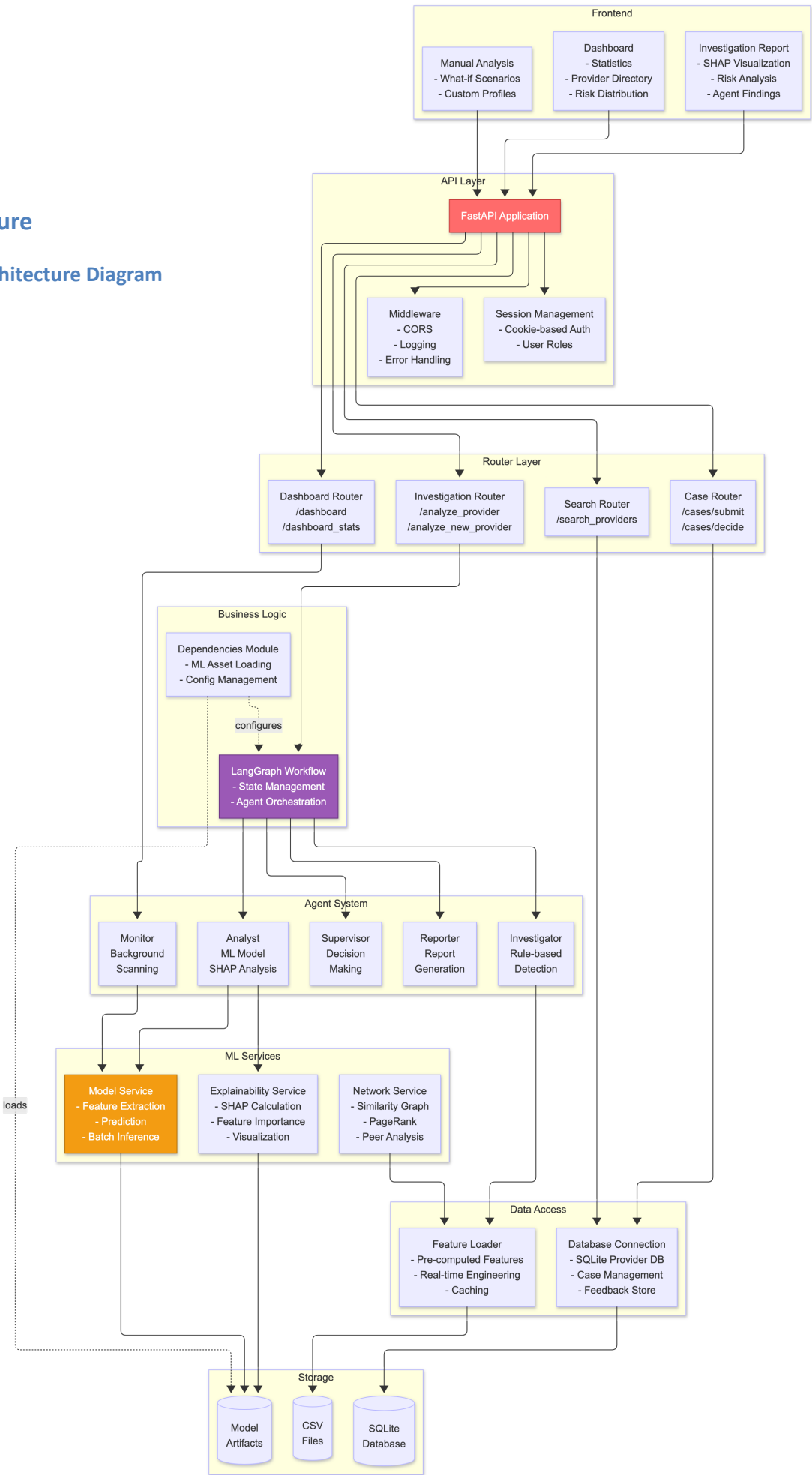
2. ARCHITECTURE DIAGRAMS

High-Level Architecture

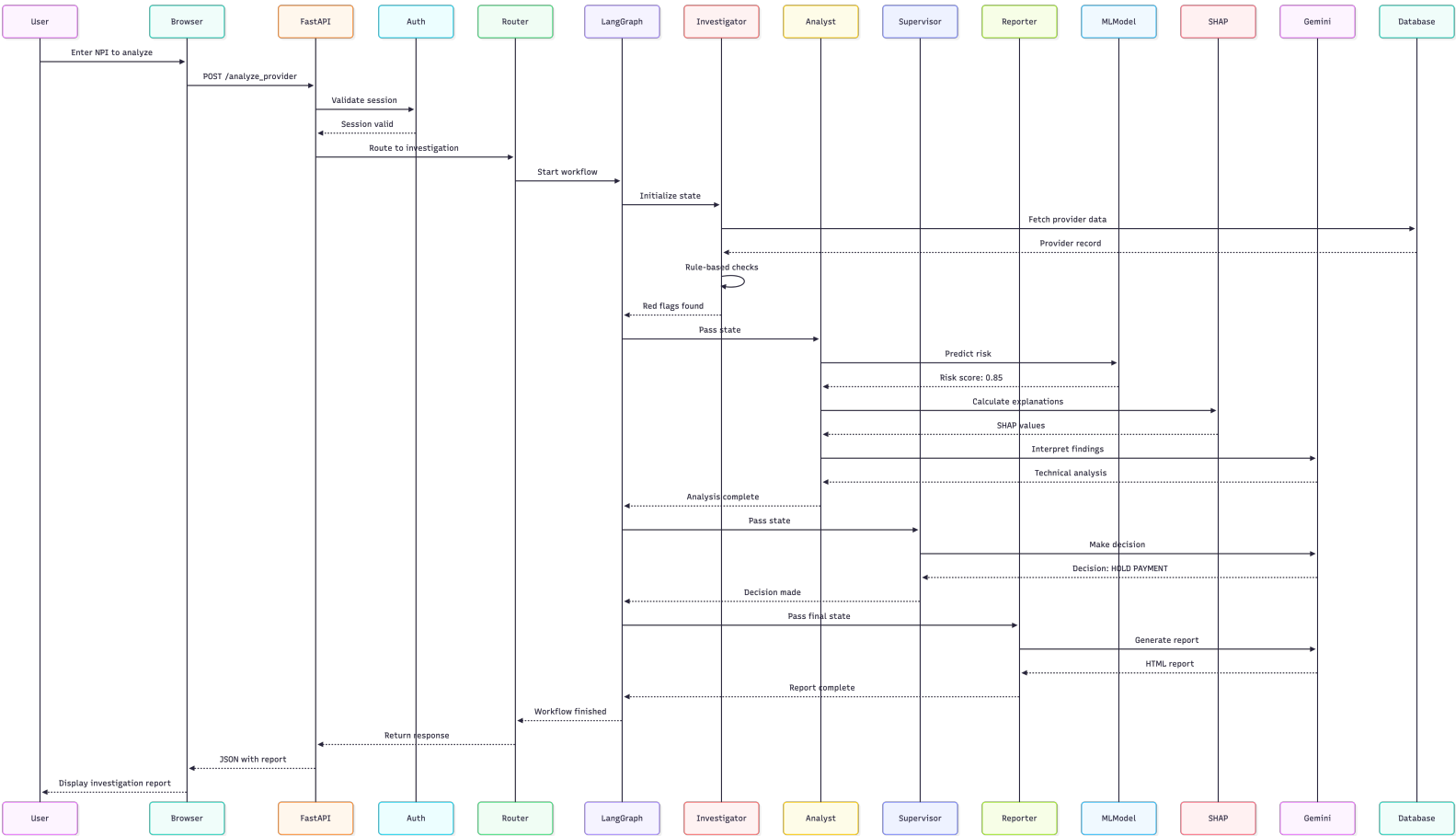


System Architecture

Detailed System Architecture Diagram

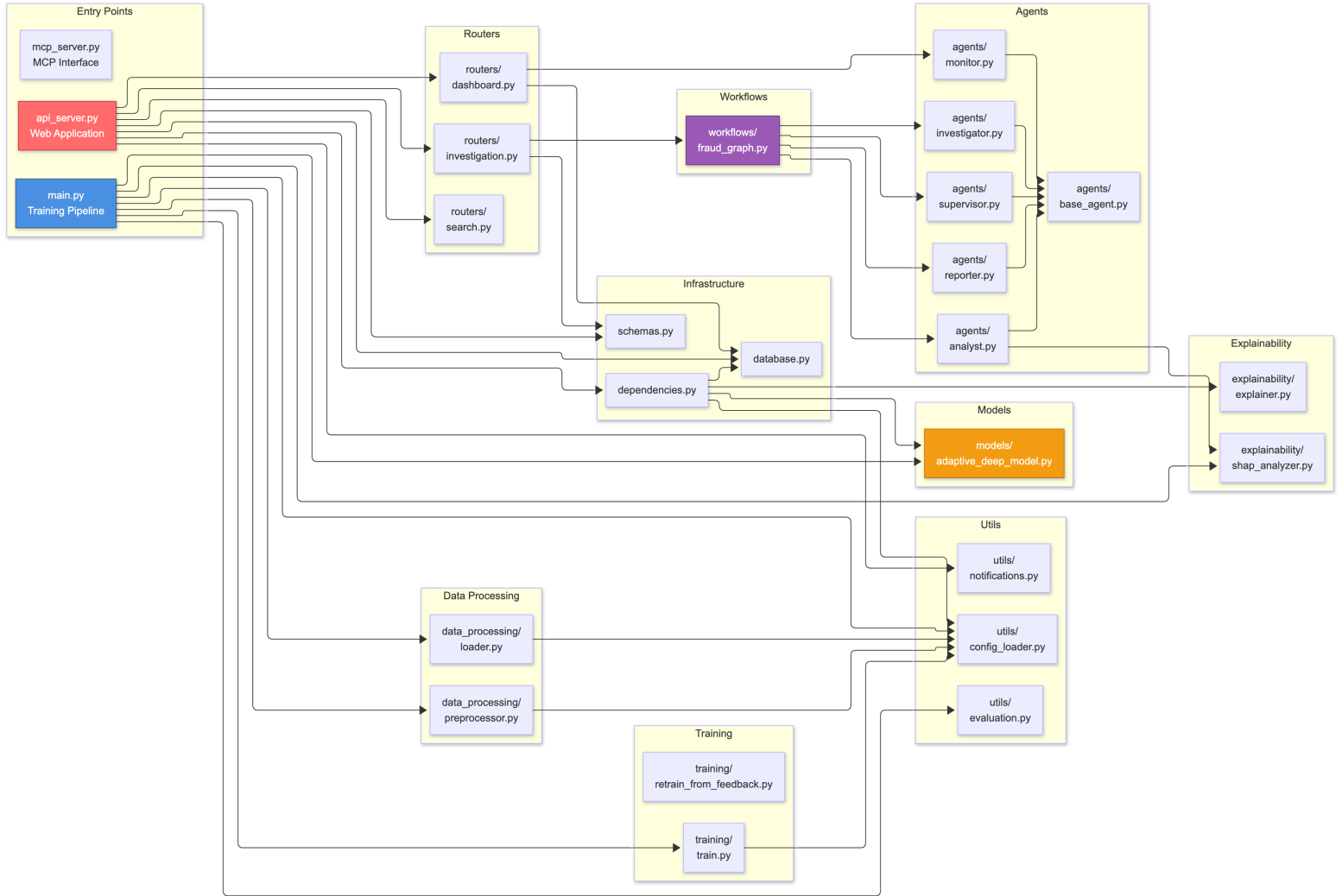


Request Flow Diagram



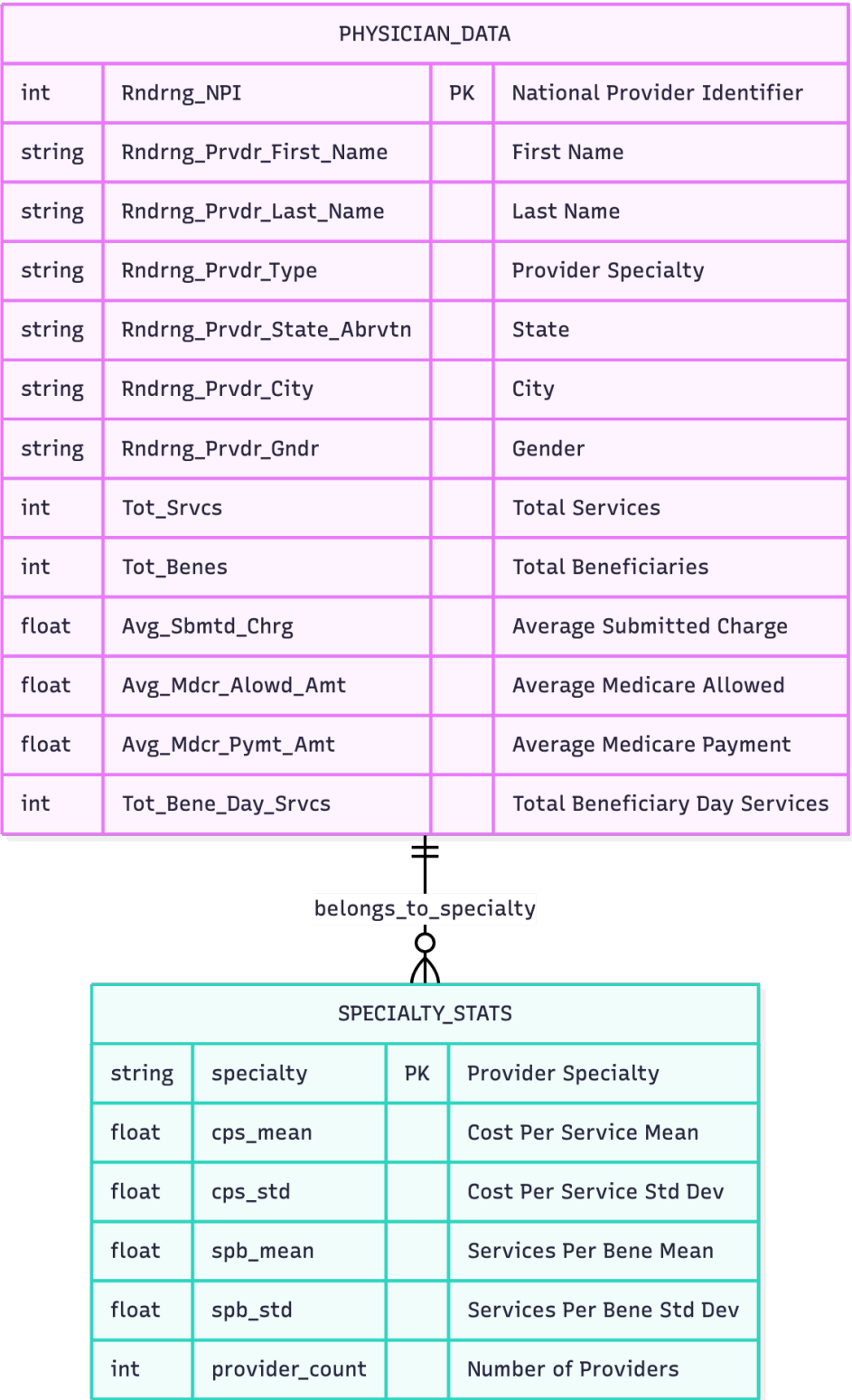
Component Interaction

Component Dependency Graph

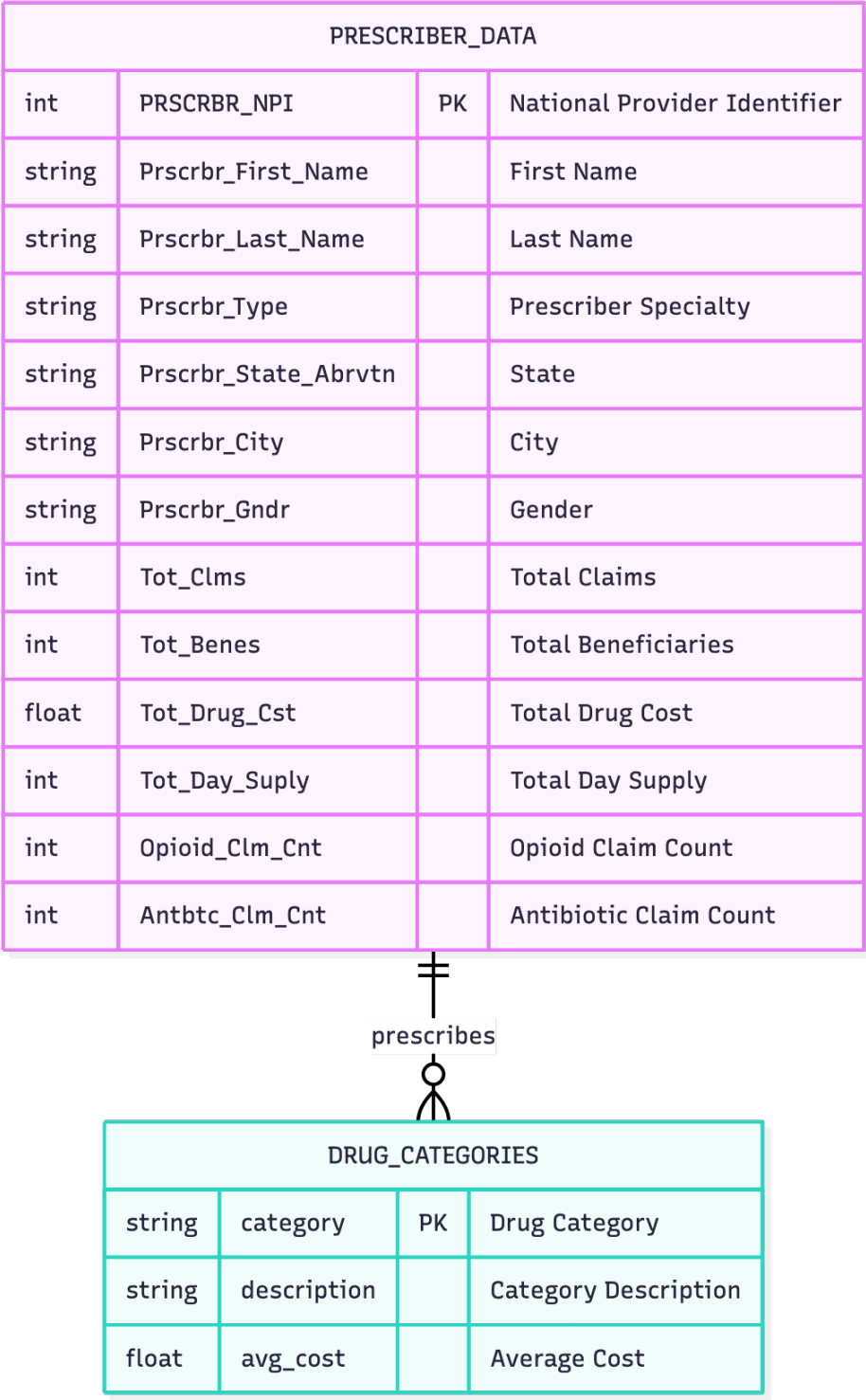


Data Model Diagrams

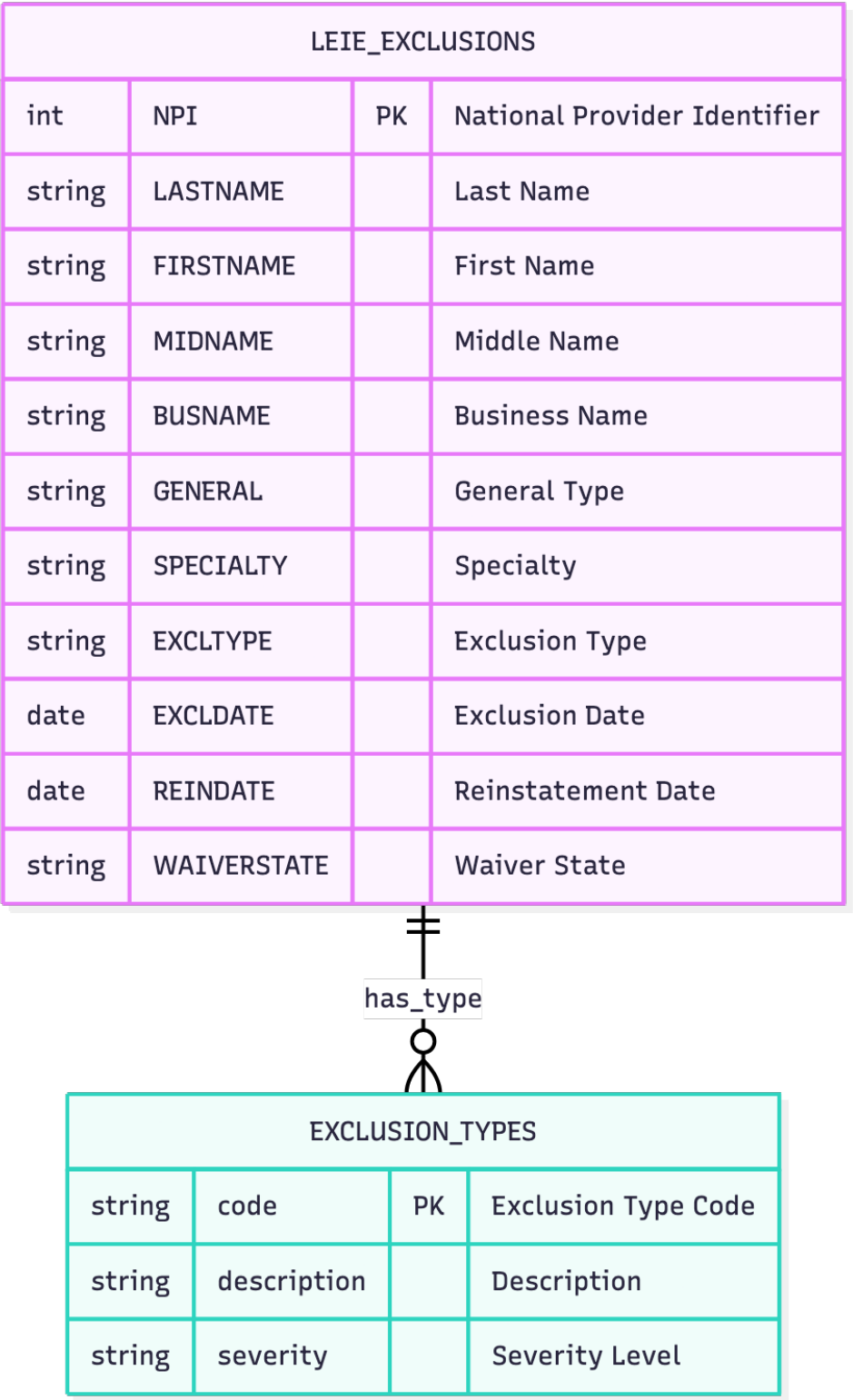
ER Diagram - Physician Dataset



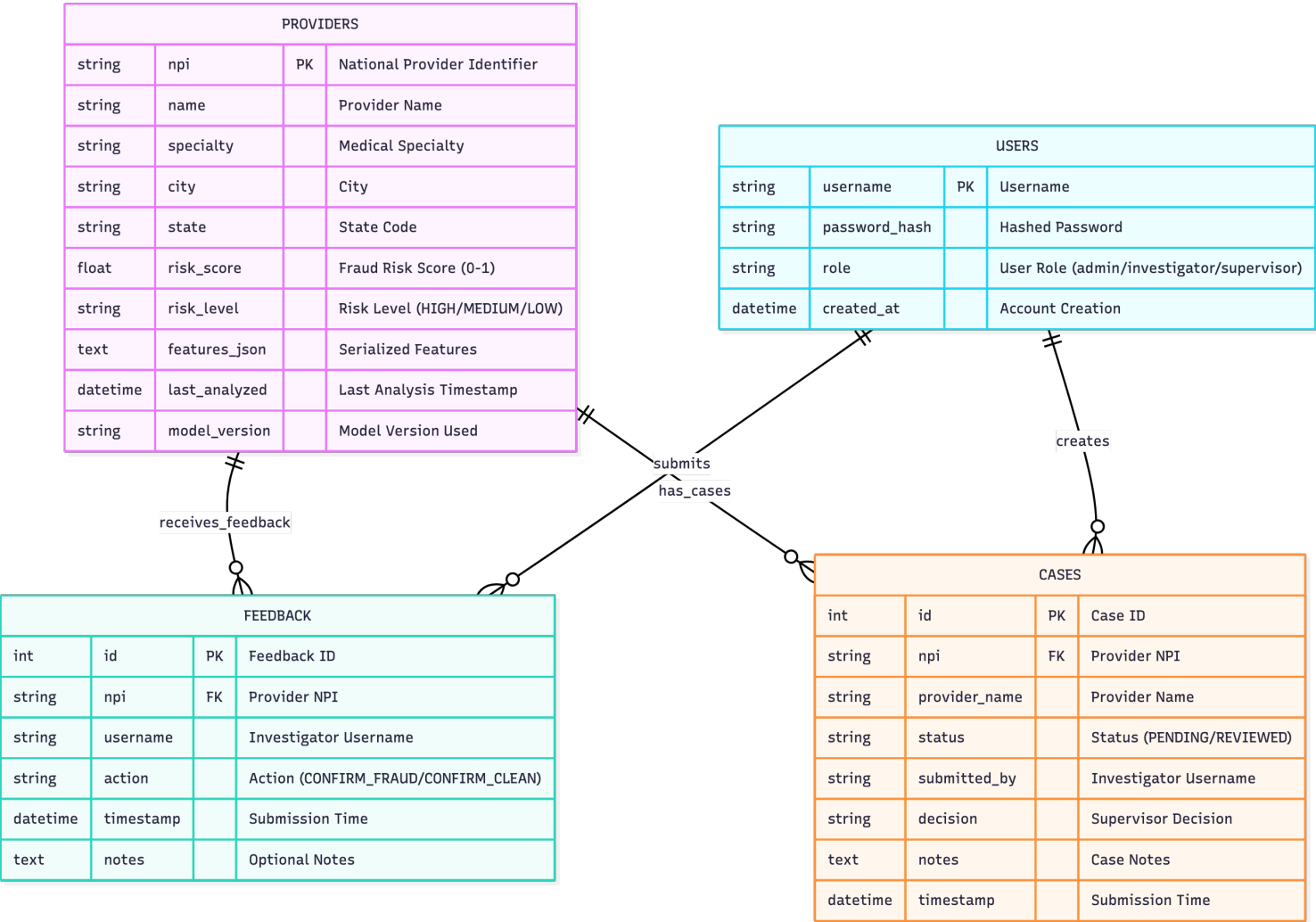
ER Diagram - Prescriber Dataset



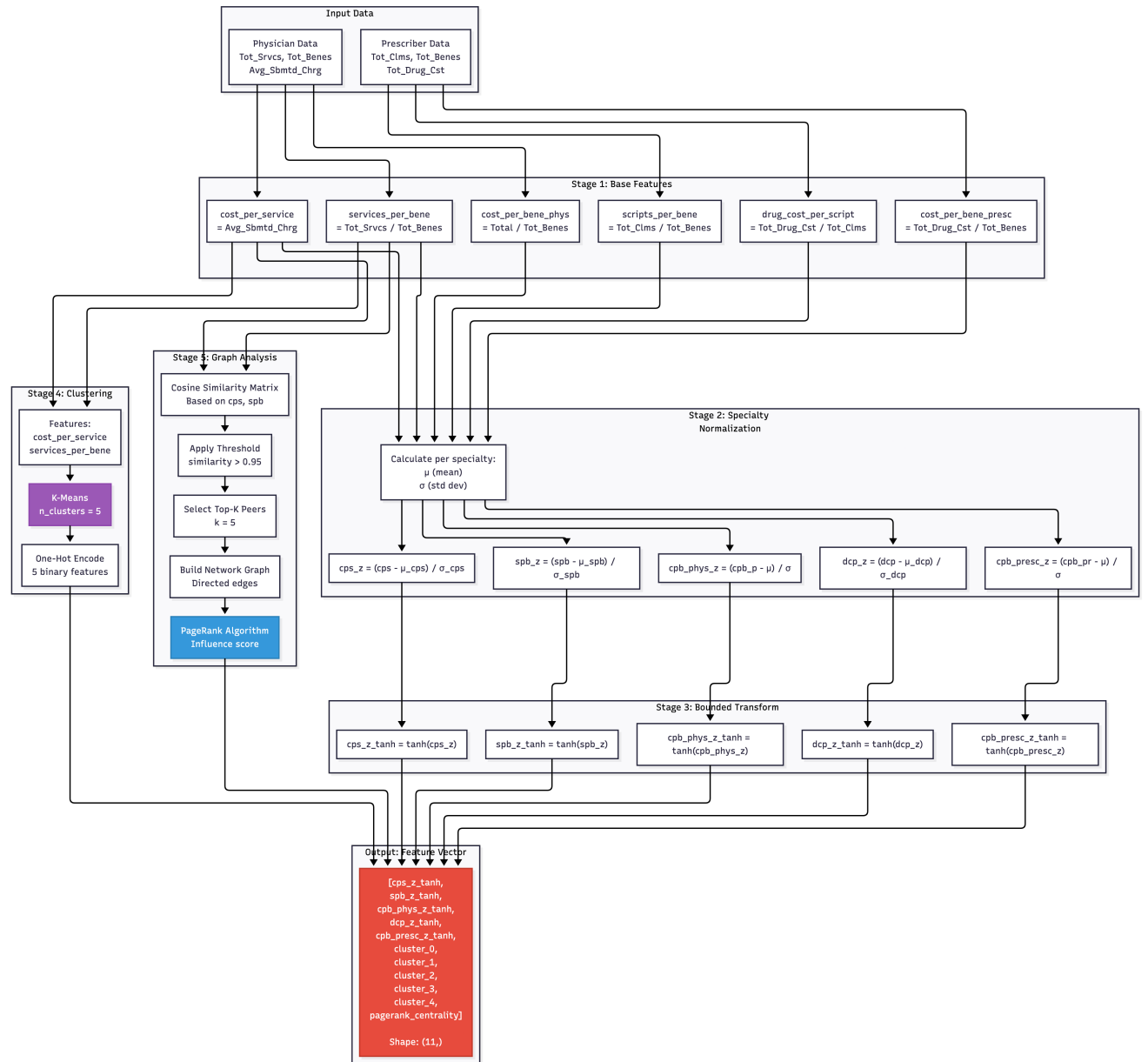
ER Diagram - LEIE Dataset



Application Database Schema

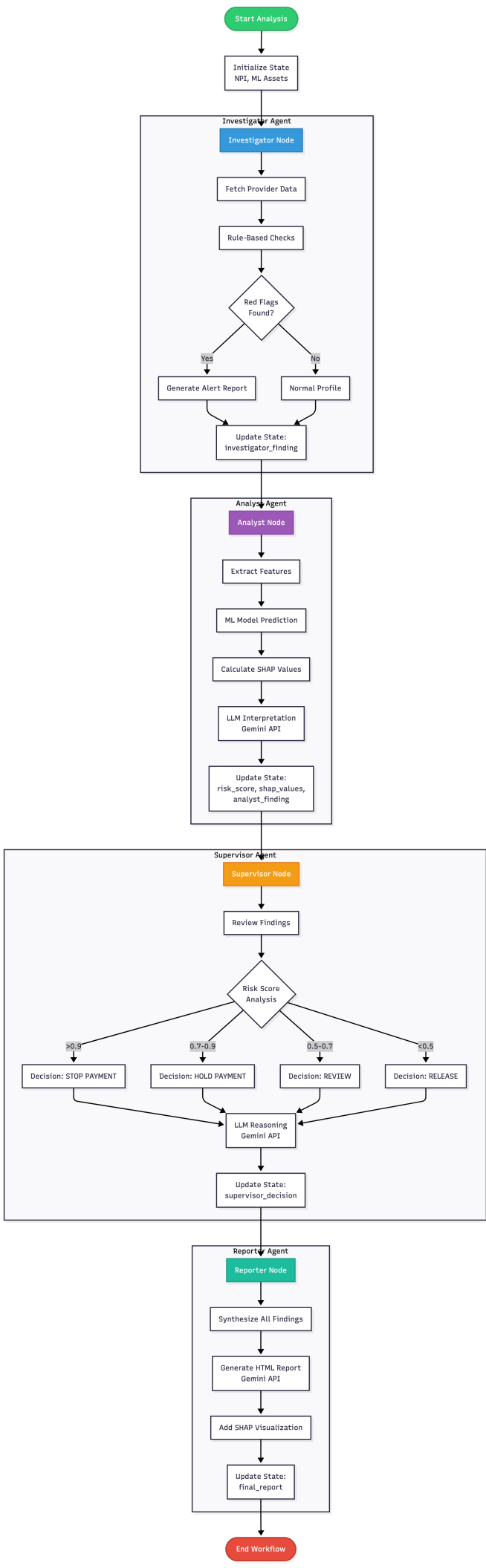


Feature Engineering Data Flow

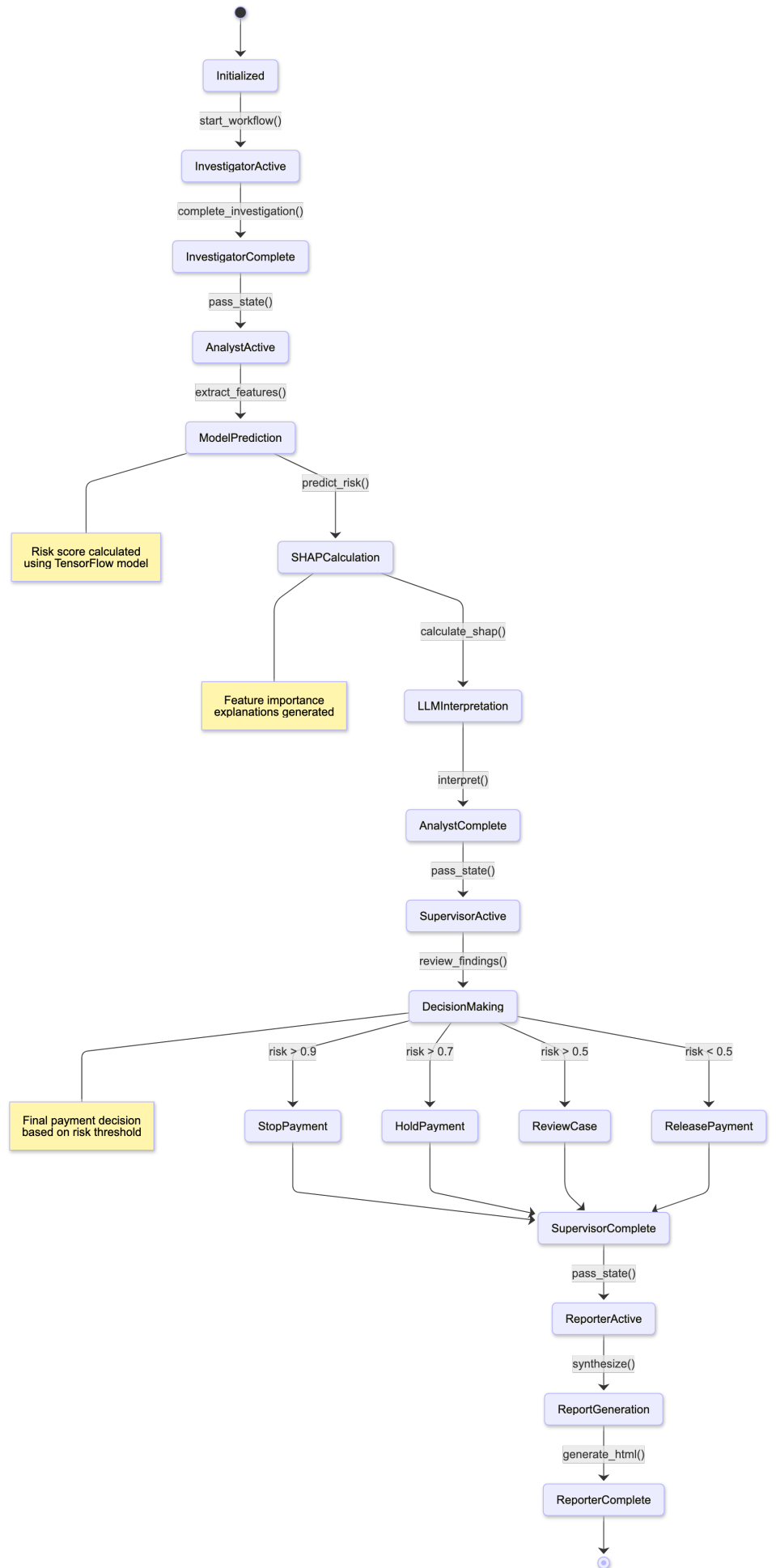


Multi-Agent Flow

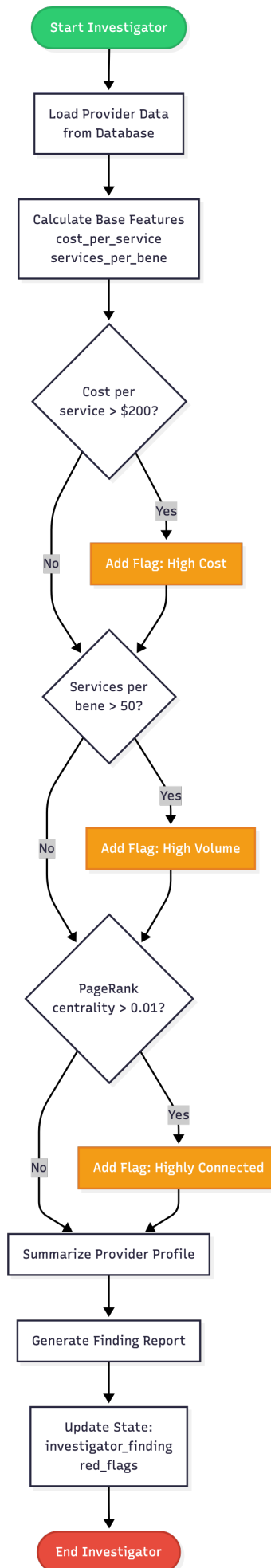
LangGraph Workflow Diagram



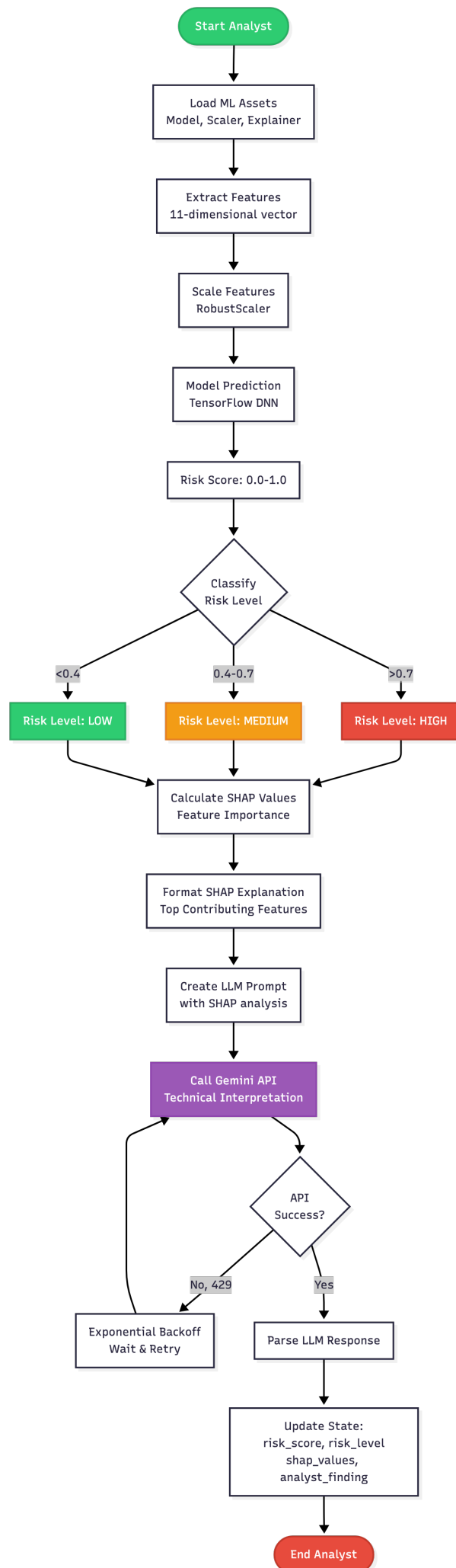
Agent State Transitions



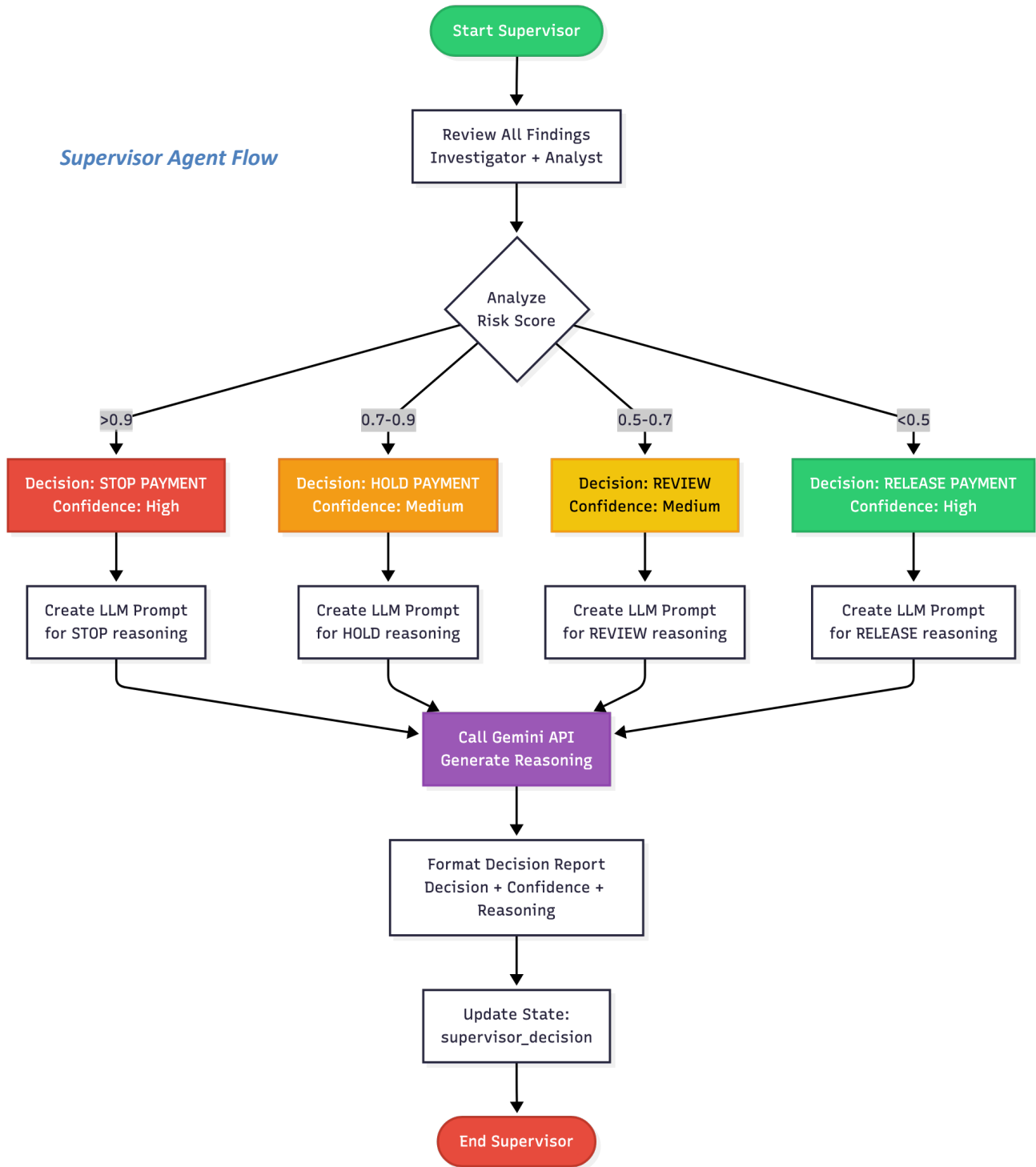
Investigator Agent Flow



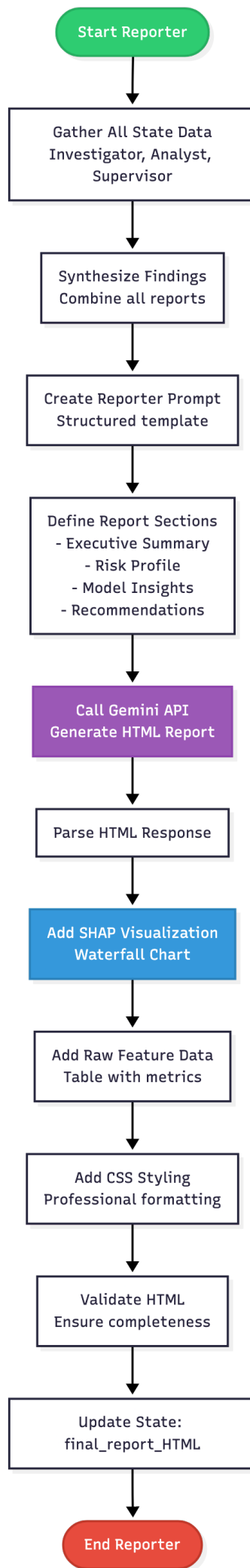
Analyst Agent Flow



Supervisor Agent Flow

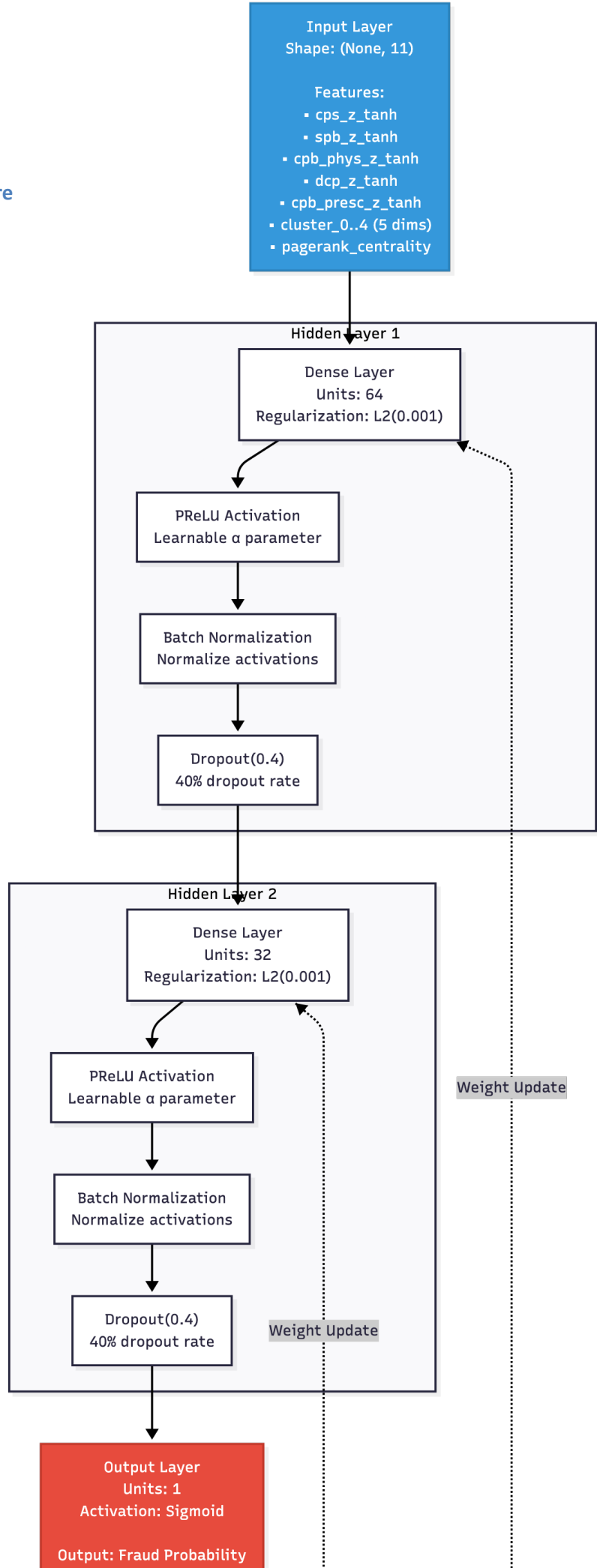


Reporter Agent Flow



Model Architecture

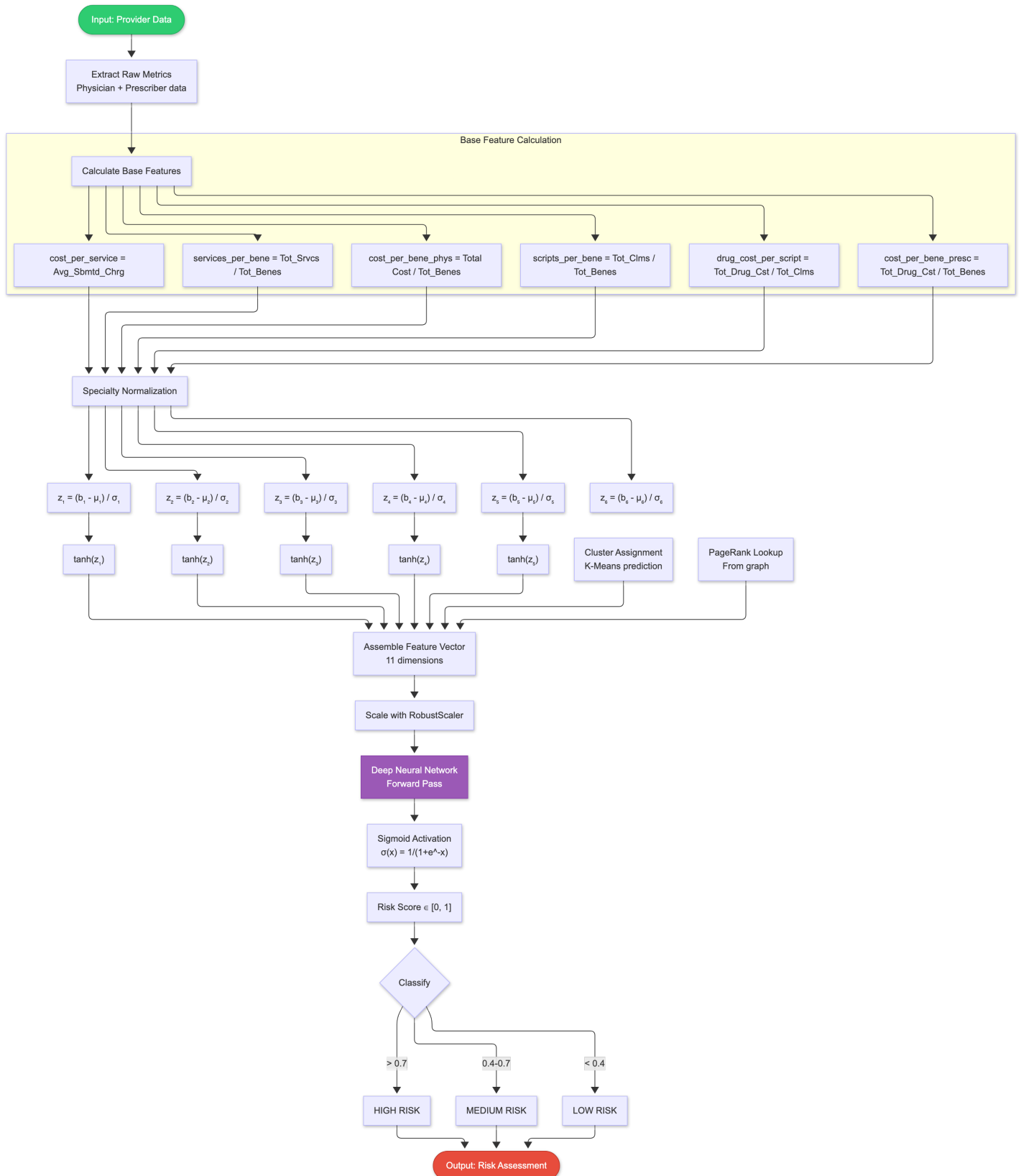
Deep Neural Network Architecture



Algorithm Diagrams and Explanations

Risk Calculation Algorithm - Detailed Explanation

Calculate a fraud risk score (0-1) for a healthcare provider based on billing patterns.



Step-by-Step Process:

1. Extract Raw Metrics from source data:

- Physician data: `Tot\_Srvcs`, `Tot\_Benes`, `Avg\_Sbmtld\_Chrg`
- Prescriber data: `Tot\_Clms`, `Tot\_Benes`, `Tot\_Drug\_Cst`

2. Calculate 6 Base Features:

- $\text{cost_per_service} = \text{Avg_Sbmtld_Chrg}$
- $\text{services_per_bene} = \text{Tot_Srvcs} / \text{Tot_Benes}$
- $\text{cost_per_bene_phys} = (\text{Avg_Sbmtld_Chrg} \times \text{Tot_Srvcs}) / \text{Tot_Benes}$
- $\text{scripts_per_bene} = \text{Tot_Clms} / \text{Tot_Benes}$
- $\text{drug_cost_per_script} = \text{Tot_Drug_Cst} / \text{Tot_Clms}$
- $\text{cost_per_bene_presc} = \text{Tot_Drug_Cst} / \text{Tot_Benes}$

These ratios normalize billing across providers of different sizes.

3. Specialty Normalization (Z-Score Calculation):

$$z = (\text{value} - \text{specialty_mean}) / \text{specialty_std}$$

Example: Internal Medicine physician with $\text{cost_per_service} = \150

- Specialty mean = \$100
- Specialty std dev = \$25
- $z = (150 - 100) / 25 = 2.0$ (2 standard deviations above average)

Compares each provider to their specialty peers (cardiologists expected to cost more than family doctors).

4. Tanh Transform (Bounded Normalization):

$$z_{\text{tanh}} = \tanh(z)$$

Range: [-1, 1]

Prevents extreme outliers from dominating the model. A provider 10 std devs above mean becomes ~1.0 instead of 10.0.

5. Cluster Assignment (K-Means):

- Assigns provider to 1 of 5 archetypes based on cost and volume
- One-hot encoded as 5 binary features

Captures practice patterns (e.g., high-cost specialist vs. high-volume routine care).

6. PageRank Centrality:

- Measures provider's position in similarity network
- High PageRank = similar to many other providers (central behavior)
- Low PageRank = unique practice patterns (isolated)

Identifies referral networks and potential collusion patterns.

7. Assemble 11-Dimensional Feature Vector:

[cps_z_tanh, spb_z_tanh, cpb_phys_z_tanh, dcp_z_tanh, cpb_presc_z_tanh,
cluster_0, cluster_1, cluster_2, cluster_3, cluster_4,
pagerank_centrality]

8. Scale Features with RobustScaler:

- Centers features around median (robust to outliers)
- Scales based on interquartile range

9. Neural Network Forward Pass:

- Layer 1: 11 → 64 neurons (feature extraction)
- Layer 2: 64 → 32 neurons (abstract representation)
- Output: 32 → 1 neuron (fraud probability)

10. Sigmoid Activation:

$$\text{risk_score} = 1 / (1 + e^{(-x)})$$

Converts network output to probability in [0, 1].

11. Classification:

- $\text{risk} > 0.7$: HIGH
- $0.4 \leq \text{risk} \leq 0.7$: MEDIUM
- $\text{risk} < 0.4$: LOW

Example End-to-End:

- Provider bills \$300/service (specialty avg: \$100, std: \$50)

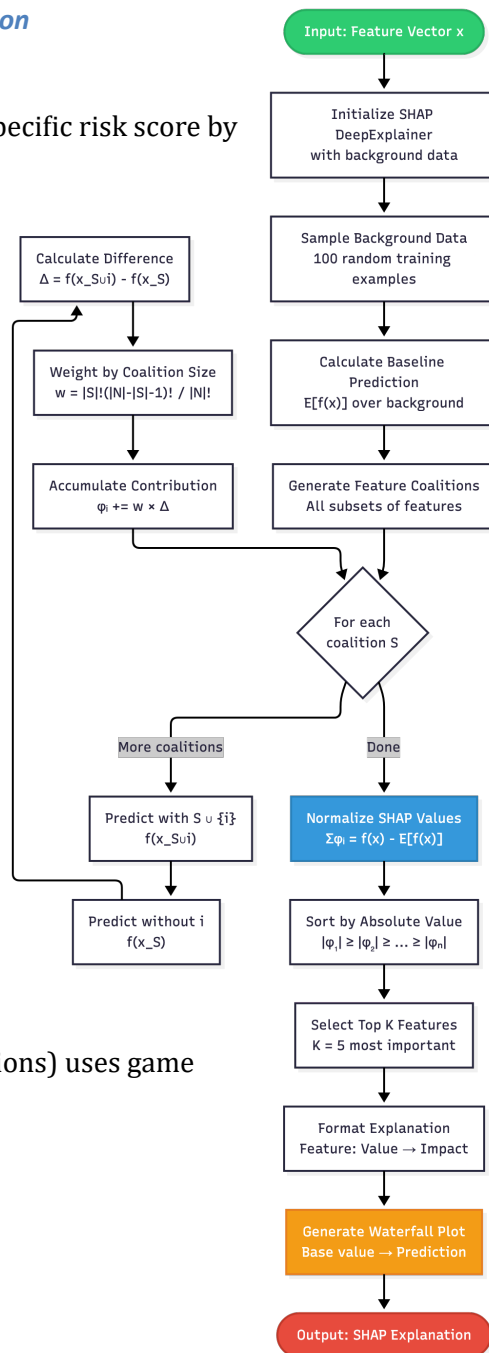
- z-score: $(300-100)/50 = 4.0$ (very high!)
- $\tanh(4.0) = 0.9993$
- After neural network: risk_score = 0.92
- Classification: HIGH RISK

SHAP Explanation Algorithm - Detailed Explanation

Purpose: Explain *why* the model assigned a specific risk score by calculating each feature's contribution.

The Problem: A neural network gives a risk score, but doesn't explain which features drove that score.

The Solution: SHAP (SHapley Additive exPlanations) uses game theory to fairly distribute credit.



Key Concept: Shapley values from cooperative game theory.

- Players: Features (cost_per_service, services_per_bene, etc.)
- Game: Predicting fraud risk
- Payout: Final prediction - baseline prediction
- Question: How much should each feature be "paid" for its contribution?

Step-by-Step Process:

1. Initialize Background Dataset:

- Sample 100 random training examples
- These represent "typical" provider profiles

2. Calculate Baseline Prediction:

$E[f(x)]$ = average prediction over background data

Example: Baseline = 0.12 (12% average risk across all providers)

3. For Each Feature i :

- Generate all possible subsets of other features (called "coalitions")
- Example: For 3 features {A, B, C}, coalitions for feature A:
- {} (no features)
- {B}
- {C}
- {B, C}

4. Calculate Marginal Contribution:

For each coalition S :

$$\Delta = f(S \cup \{i\}) - f(S)$$

Meaning: Prediction with feature i included - prediction without

Example:

- $f(\{\text{cost, volume}\}) = 0.7$
- $f(\{\text{cost, volume, centrality}\}) = 0.85$
- $\Delta = 0.85 - 0.7 = 0.15$ (centrality adds +0.15 to risk)

5. Weight by Coalition Size:

$$\text{weight} = |S|! \times (n - |S| - 1)! / n!$$

Gives more weight to coalitions of medium size

- Small coalitions (few features): Less reliable
- Large coalitions (many features): Less informative about feature i

6. Accumulate Weighted Contributions:

$$\varphi_i = \Sigma [\text{weight} \times \Delta]$$

Result: Feature i's SHAP value

7. Normalization Property:

$$\Sigma \varphi_i = f(x) - E[f(x)]$$

Example: If prediction = 0.85 and baseline = 0.12:

- Total to distribute: $0.85 - 0.12 = 0.73$
- SHAP values sum to 0.73

8. Interpret SHAP Values:

$$\text{Prediction} = \text{Baseline} + \text{SHAP}_1 + \text{SHAP}_2 + \dots + \text{SHAP}_n$$

$$= 0.12 + 0.40 + 0.25 + \dots + 0.08$$

$$= 0.85$$

Example SHAP Breakdown:

Feature	Value	SHAP	Interpretation
Baseline		0.12	Average provider
cost_per_service	\$300	+0.40	Very high cost (+40%)
services_per_bene	60	+0.25	High volume (+25%)
cluster_1 (High Risk)	Yes	+0.10	Risky archetype (+10%)
pagerank_centrality	0.02	-0.02	Not well-connected (-2%)
Final Prediction		0.85	85% fraud risk

Visualization (Waterfall Chart):

0.12 (Baseline)

→ +0.40 (Cost pushed right)

→ +0.25 (Volume pushed right)

→ +0.10 (Cluster pushed right)

→ -0.02 (Centrality pulled left)

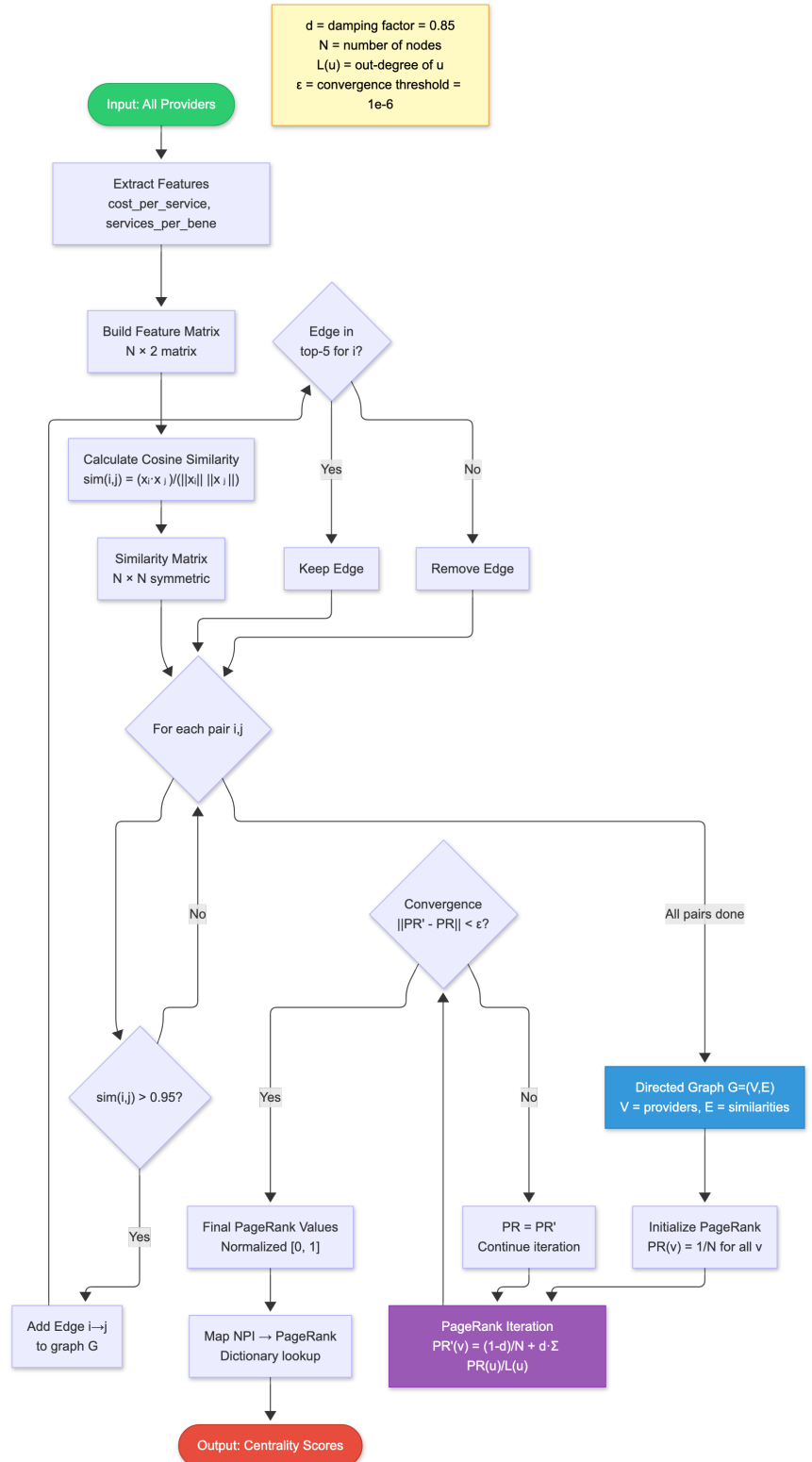
= 0.85 (Final)

Network Analysis (PageRank) - Detailed Explanation

Purpose: Identify providers who are central in the similarity network (potential fraud rings or referral schemes).

Analogy: Google's PageRank for web pages, applied to provider similarity.

Key Idea: A provider is "important" if similar to many other providers who are themselves important.



Step-by-Step Process:

1. Extract Features for all providers:

- cost_per_service
- services_per_bene

2. Build N×2 Feature Matrix:

[provider_1]: [100, 20]

[provider_2]: [105, 22]

...

[provider_N]: [95, 18]

3. Calculate Cosine Similarity between all pairs:

$$\text{sim}(i, j) = (\mathbf{x}_i \cdot \mathbf{x}_j) / (||\mathbf{x}_i|| \times ||\mathbf{x}_j||)$$

Example: Providers A=[100, 20] and B=[200, 40]:

- Dot product: $100 \times 200 + 20 \times 40 = 20,800$
- Magnitude A: $\sqrt{(100^2 + 20^2)} = 101.98$
- Magnitude B: $\sqrt{(200^2 + 40^2)} = 203.96$
- Similarity: $20,800 / (101.98 \times 203.96) = 0.999$ (nearly identical!)

4. Apply Threshold (0.95):

- Only keep edges where similarity > 0.95
- Rationale: Focus on very similar providers

5. Select Top-K (k=5):

- For each provider, keep only 5 most similar peers
- Why: Prevents highly connected hubs from dominating

6. Build Directed Graph:

- Nodes: Providers
- Edges: Provider $i \rightarrow j$ if j is in top-5 most similar to i

7. Initialize PageRank:

$$\text{PR}(v) = 1/N \text{ for all providers}$$

Example: 10,000 providers $\rightarrow$ $PR(v) = 0.0001$ initially

8. Iterate PageRank Formula:

$$PR'(v) = (1-d)/N + d \times \sum [PR(u) / L(u)]$$

Variables:

- $d = 0.85$: Damping factor (probability of following a link)
- N : Total providers
- $PR(u)$: Current PageRank of provider u
- $L(u)$: Number of outgoing links from u
- Σ : Sum over all u that link to v

Intuition:

- 15% chance: Random jump to any provider $(1-d)/N$
- 85% chance: Follow a link from similar provider

9. Converge:

- Repeat until $||PR' - PR|| < 1e-6$
- Typically 30-50 iterations

10. Interpret Results:

- High PageRank (> 0.01): Hub provider with many similar peers
- Could indicate referral network
- Could indicate standard practice in specialty
- Low PageRank (< 0.001): Isolated provider with unique behavior
- Could be specialized niche
- Could be anomalous billing

Example Scenario:

Network:

Provider A $\rightarrow$ similar to B, C, D, E, F (total 5)
Provider B $\rightarrow$ similar to A, G, H (total 3)
Provider C $\rightarrow$ similar to A, I (total 2)
...

After PageRank:

A: $PR = 0.025$ (high centrality, well-connected hub)

```

B: PR = 0.012 (moderate)
C: PR = 0.008 (moderate)
...

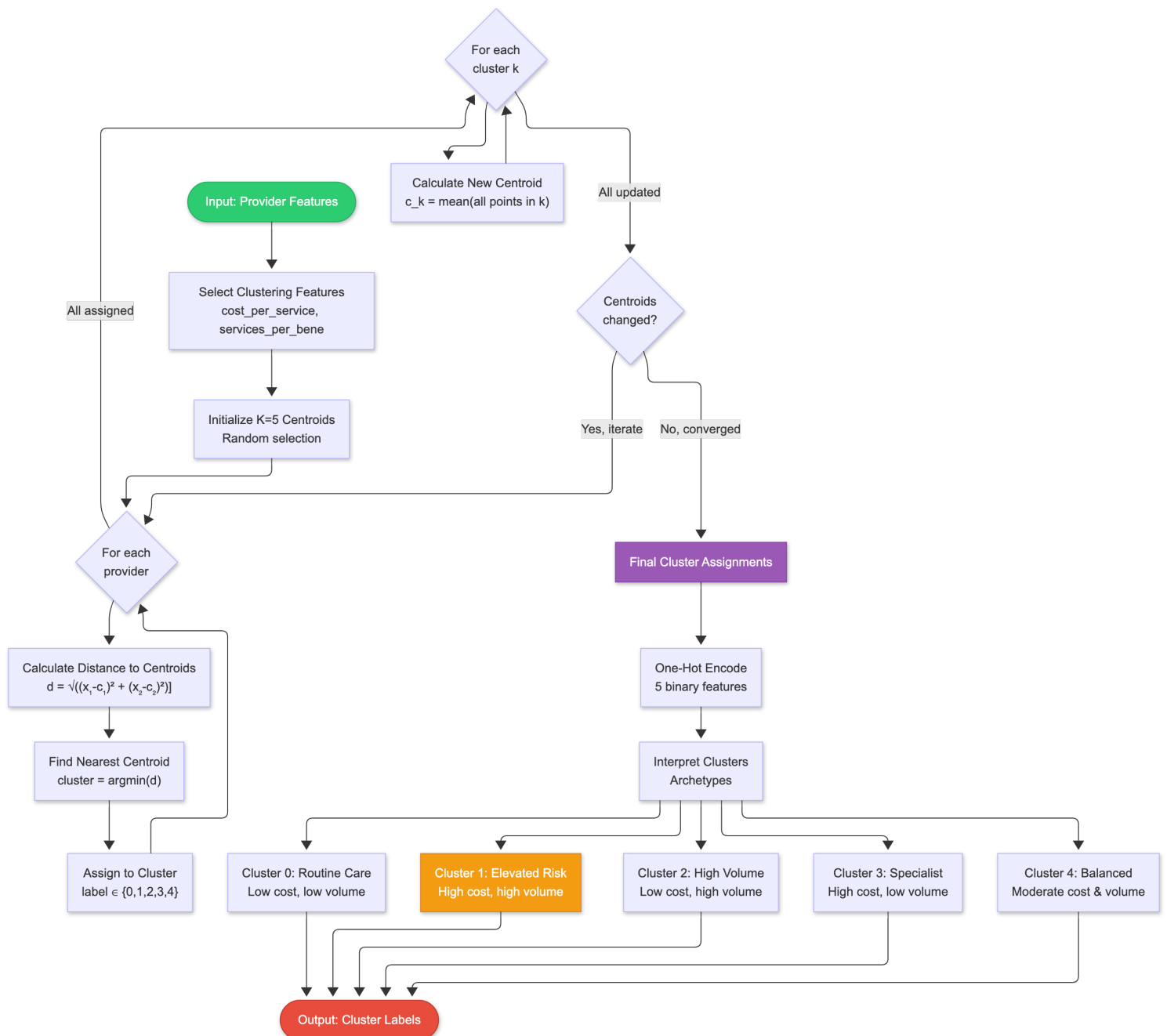
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Fraud Detection Use:

- High centrality + high risk score → Potential fraud ring leader
- Low centrality + high risk score → Isolated bad actor

K-Means Clustering - Detailed Explanation

Purpose: Group providers into 5 behavioral archetypes based on cost and volume patterns.



Why Clustering?:

- Captures non-linear patterns
- Creates interpretable archetypes
- Adds categorical features to model

Step-by-Step Process:

24. Select Features:

- cost_per_service (x-axis)
- services_per_bene (y-axis)

2. Initialize 5 Centroids randomly:

$C_1 = [50, 10]$ (low cost, low volume)

$C_2 = [300, 80]$ (high cost, high volume)

$C_3 = [60, 70]$ (low cost, high volume)

$C_4 = [250, 15]$ (high cost, low volume)

$C_5 = [150, 40]$ (moderate, moderate)

3. Assignment Step:

For each provider, calculate distance to each centroid:

$$d = \sqrt{[(x - c_x)^2 + (y - c_y)^2]}$$

Assign to nearest centroid.

Example: Provider at [100, 25]

- $d(C_1) = \sqrt{[(100-50)^2 + (25-10)^2]} = \sqrt{2725} = 52.2$
- $d(C_2) = \sqrt{[(100-300)^2 + (25-80)^2]} = \sqrt{43025} = 207.5$
- $d(C_3) = \sqrt{[(100-60)^2 + (25-70)^2]} = \sqrt{3625} = 60.2$
- $d(C_4) = \sqrt{[(100-250)^2 + (25-15)^2]} = \sqrt{22600} = 150.3$
- $d(C_5) = \sqrt{[(100-150)^2 + (25-40)^2]} = \sqrt{2725} = 52.2$
- Nearest: C_1 or C_5 (tie $\rightarrow$ choose C_1)

4. Update Step:

For each cluster k , calculate new centroid as mean of all assigned points:

$$c_k = (1/|k|) \times \sum[\text{all points in cluster } k]$$

5. Convergence Check:

- If centroids didn't move: STOP
- Else: Repeat assignment and update

6. Final Cluster Interpretation:

Cluster 0: Routine Care

- Low cost (\$50-100/service)
- Low volume (10-30 services/bene)
- Examples: Primary care, preventive medicine

Cluster 1: Elevated Risk Profile

- High cost (\$200+/service)
- High volume (60+ services/bene)
- FRAUD INDICATOR: Unusual to have both high

Cluster 2: High Volume Operators

- Low cost (\$50-80/service)
- High volume (60+ services/bene)
- Examples: Urgent care, walk-in clinics

Cluster 3: Specialist Care

- High cost (\$200+/service)
- Low volume (5-20 services/bene)
- Examples: Surgical specialists, rare procedures

Cluster 4: Balanced Practice

- Moderate cost (\$100-150/service)
- Moderate volume (30-50 services/bene)
- Examples: General specialists

7. One-Hot Encoding:

Convert cluster assignment to 5 binary features:

Provider in Cluster 1 $\rightarrow [0, 1, 0, 0, 0]$

Provider in Cluster 3 $\rightarrow [0, 0, 0, 1, 0]$

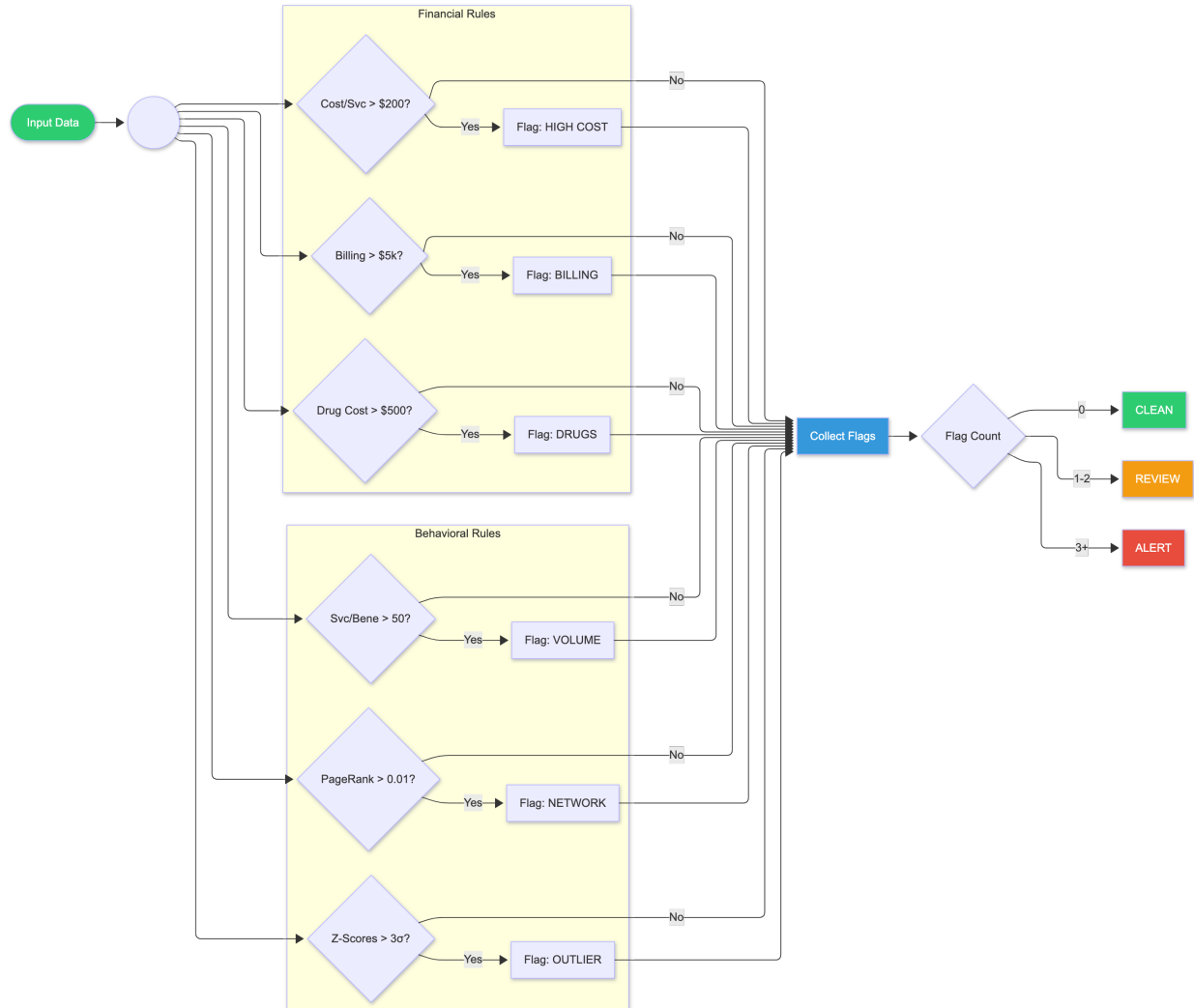
Model Usage:

- Neural network learns that Cluster 1 is risky
- Cluster 3 is legitimate but expensive
- Provides context missing from raw cost/volume alone

Anomaly Detection - Detailed Explanation

Purpose: Rule-based initial screening before ML model.

Quick, interpretable red flags that don't require complex computation.



7-Rule System:

Rule 1: Excessive Cost

IF cost_per_service > \$200 THEN FLAG: HIGH

Rationale: Medicare average is ~\$100. Above \$200 suggests upcoding or unnecessary procedures.

Rule 2: High Volume

IF services_per_bene > 50 THEN FLAG: MEDIUM

Rationale: Average is 20-30. Above 50 suggests overutilization or billing for services not rendered.

Rule 3: Excessive Billing

IF cost_per_bene > \$5000 THEN FLAG: HIGH

Rationale: Annual Medicare spending averages \$3000/beneficiary. Much higher suggests fraud.

Rule 4: Expensive Drugs

IF drug_cost_per_script > \$500 THEN FLAG: MEDIUM

Rationale: Median prescription cost is \$50-100. High-cost drugs should be rare.

Rule 5: Highly Connected

IF pagerank Centrality > 0.01 THEN FLAG: LOW

Rationale: Very connected providers may indicate referral rings. Needs investigation.

Rule 6: Cost Outlier

IF cps_z_tanh > 0.8 THEN FLAG: HIGH

Rationale: $\tanh(3) \approx 0.995$, so 0.8 means ~2 std devs above specialty mean. Statistical outlier.

Rule 7: Volume Outlier

IF spb_z_tanh > 0.8 THEN FLAG: HIGH

Rationale: Same as Rule 6, for volume instead of cost.

Severity Classification:

- 0 flags: CLEAN (no investigation needed)
- 1-2 flags: REVIEW (minor anomalies, worth checking)
- 3+ flags: ALERT (multiple red flags, priority investigation)

Example:

Provider X:

<i>cost_per_service = \$350</i>	<i>→ Rule 1: HIGH</i>
<i>services_per_bene = 65</i>	<i>→ Rule 2: MEDIUM</i>
<i>cost_per_bene = \$6,500</i>	<i>→ Rule 3: HIGH</i>
<i>drug_cost_per_script = \$120</i>	<i>→ Rule 4:</i>
<i>pagerank = 0.005</i>	<i>→ Rule 5:</i>
<i>cps_z_tanh = 0.92</i>	<i>→ Rule 6: HIGH</i>
<i>spb_z_tanh = 0.75</i>	<i>→ Rule 7:</i>

Total Flags: 4 → STATUS: ALERT

Integration with ML:

- Anomaly detection runs BEFORE ML model
- Provides context for investigators
- ML model may override (e.g., specialist with legitimately high cost)
- Both used together for final decision

Component Descriptions

Presentation Layer

- Dashboard: Main UI for viewing provider statistics, risk scores, and analytics
- Investigation Reports: Detailed HTML reports with SHAP visualizations and risk analysis
- Search Interface: Advanced search and filter capabilities for provider directory

Application Layer

- FastAPI Server: High-performance async REST API with automatic documentation
- Session Management: HTTP-only cookie-based authentication
- Request Validation: Pydantic models for request/response validation

Agent Layer

- Investigator Agent: Performs initial rule-based anomaly detection
- Analyst Agent: Runs ML model and generates SHAP explanations
- Supervisor Agent: Makes final payment decisions based on risk analysis
- Reporter Agent: Synthesizes findings into narrative reports
- LangGraph Engine: Orchestrates multi-agent workflows with state management

Service Layer

- ML Model: Deep neural network for fraud risk prediction
- SHAP Explainer: Generates feature importance explanations
- Network Analysis: Cosine similarity-based peer comparison

Data Layer

- SQLite Database: Stores provider records, cases, and feedback
- CSV Files: Raw Medicare datasets (Physician, Prescriber, LEIE)
- Feature Store: Pre-computed engineered features for fast inference

Technology Stack

Backend Technologies

Core Framework

- FastAPI 0.95+: Modern async web framework
- Auto-generated OpenAPI documentation
- Built-in request validation
- High performance (comparable to Node.js/Go)

Machine Learning

- TensorFlow 2.10+: Deep learning framework
- Custom focal loss for imbalanced datasets
- Model serialization and versioning
- scikit-learn 1.0+: Traditional ML and preprocessing
- RobustScaler for outlier handling
- KMeans clustering for archetype detection
- SHAP 0.41+: Model explainability
- TreeExplainer for tree-based models
- DeepExplainer for neural networks
- imbalanced-learn: SMOTE for class imbalance

AI Orchestration

- LangGraph 0.0.26+: Workflow engine for multi-agent systems
- State management across agent interactions
- Conditional routing based on analysis results
- LangChain Core: Foundation for LLM integration

Data Processing

- Pandas 1.5+: Data manipulation and analysis
- NumPy 1.23+: Numerical computing
- NetworkX 2.8+: Graph analysis (PageRank centrality)

Database

- SQLite3: Embedded database for provider records, cases, and feedback

- Python sqlite3 module: Native database interface

Frontend Technologies

UI Framework

- TailwindCSS 3.x: Utility-first CSS framework
- Responsive design system
- Custom color schemes
- Alpine.js: Lightweight reactive framework (optional)

Templating

- Jinja2: Server-side template rendering
- Template inheritance
- Context variables
- Custom filters

Visualization

- Chart.js 3.x: Interactive charts
- Risk distribution histograms
- Feature importance bar charts
- Time series plots
- Custom CSS/JS: SHAP waterfall visualizations

Development & Deployment

Server

- Uvicorn: ASGI server for FastAPI
- Hot reload during development
- Multi-worker support in production

Environment Management

- Python dotenv: Environment variable management
- Virtual Environment: Isolated Python dependencies

Containerization

- Docker: Container platform
- Docker Compose: Multi-container orchestration

External Services

LLM Provider

- Google Gemini Pro: Large language model for report generation
- API-based integration
- Rate limiting and retry logic

Data Sources

- CMS Public Datasets: Medicare claims data
- OIG LEIE: List of Excluded Individuals/Entities

Contribution of Candidate

Significantly enhanced the system from a basic prototype to a production-ready agentic application. Key contributions include:

- **Agentic Workflow Implementation:** Designed and coded the `FraudAnalysisState` and graph transitions in `src/workflows/fraud_graph.py`.
- **Supervisor Mode & Role-Based Access:** Implemented a secure authentication system distinguishing between “Investigators” and “Supervisors”.
- **Real-Time Performance Optimization:** Implemented an in-memory caching layer to reduce redundant computation.
- **Semantic Search Engine:** Integrated a vector-based search allowing natural language queries.
- **Operational Integration:** Added PDF Report Generation and SMTP Email Notifications.
- **UI/UX Overhaul:** Redesigned the frontend using TailwindCSS for a professional look.

Results Obtained

UnSupervised Learning

Optimal threshold from P-R curve: 0.8308

--- Classification Report (using optimal threshold) ---

	precision	recall	f1-score	support
Not Fraud	0.99	0.99	0.99	350586
Fraud	0.23	0.31	0.26	2934
accuracy				0.99 353520
macro avg				0.61 0.65 0.63 353520
weighted avg				0.99 0.99 0.99 353520

ROC AUC Score: 0.9206

UnSupervised Applying SMOTE

Optimal threshold from P-R curve: 0.8878

--- Classification Report (using optimal threshold) ---

	precision	recall	f1-score	support
Not Fraud	0.99	0.99	0.99	350586
Fraud	0.27	0.34	0.30	2934
accuracy				0.99 353520
macro avg	0.63	0.67	0.65	353520
weighted avg	0.99	0.99	0.99	353520

ROC AUC Score: 0.9159

UnSupervised Applying SMOTE and Focal Loss

Optimal threshold from P-R curve: 0.7183

--- Classification Report (using optimal threshold) ---

	precision	recall	f1-score	support
Not Fraud	0.99	0.99	0.99	350195
Fraud	0.32	0.33	0.33	3325
accuracy				0.99 353520
macro avg	0.66	0.66	0.66	353520
weighted avg	0.99	0.99	0.99	353520

ROC AUC Score: 0.9351

The combination of SMOTE and Focal Loss has resulted in the most robust model, with the highest AUC score we've seen. I have saved this model and disabled automatic retraining in the configuration.

Supervised Learning with SMOTE and Focal Loss

Successfully transitioned with configuration to test the pipeline to Supervised Learning using the LEIE dataset as ground truth.

Key Achievements:

Real Fraud Labels: The model now learns from actual confirmed fraud cases (LEIE) instead of synthetic outliers.

Handled Extreme Imbalance: SMOTE successfully balanced the training data (from ~250 fraud cases to ~1 million), allowing the model to learn effectively.

Improved Data Quality: Implemented "Coalesce" logic to fill missing specialties and added has_part_b/has_part_d flags.

Final Results (on Real LEIE Labels):

F1-Score: 0.31

Recall: 0.33

ROC-AUC: 0.9118

Significance: We achieved the same high performance on real-world fraud labels as we did on the easier synthetic tasks. This is a major validation of our approach. The model is now a robust, production-grade fraud detector trained on ground truth. Reverted back to UnSupervised learning.

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Code Implementation

Code Repository

<https://github.com/UCKalyan/HealthCareFraudDetection>

DataSets: (Publicly Available Datasets)

The project uses public datasets from the Centers for Medicare & Medicaid Services (CMS) for year 2023 (Latest).

Medicare Part D Prescriber Data: <https://www.cms.gov/Research-Statistics-Data-and-Systems/Statistics-Trends-and-Reports/Medicare-Provider-Charge-Data/Part-D-Prescriber>

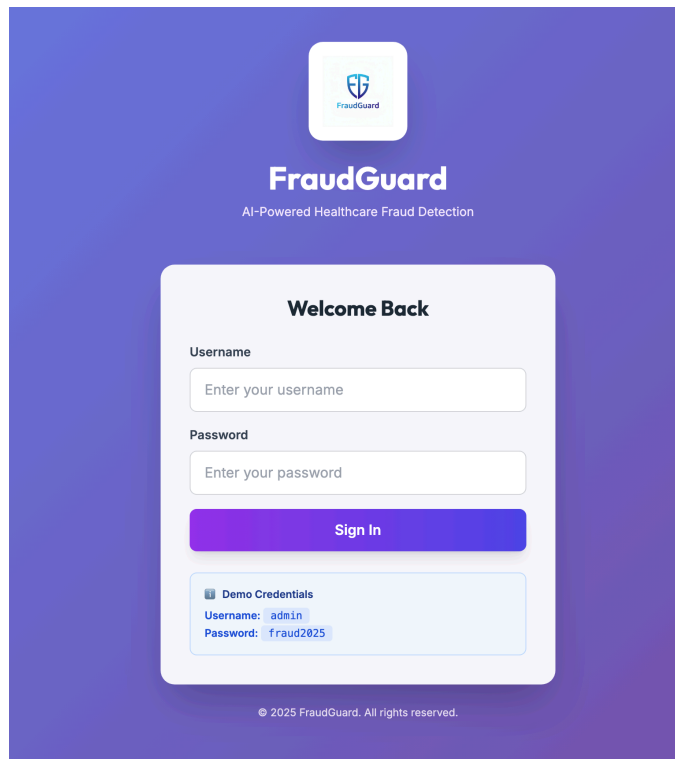
Medicare Physician & Other Supplier Data: <https://www.cms.gov/Research-Statistics-Data-and-Systems/Statistics-Trends-and-Reports/Medicare-Provider-Charge-Data/Physician-and-Other-Supplier>

LEIE (List of Excluded Individuals/Entities) - Known fraudulent providers

<https://oig.hhs.gov/exclusions/downloadables/UPDATED.csv>

UI/UX Application

Login Screen



The login screen for FraudGuard features a purple gradient background. At the top center is the FraudGuard logo, which consists of a shield icon with a checkmark and the text 'FraudGuard' below it. Below the logo, the text 'FraudGuard' is displayed in a large, bold, white font, followed by the tagline 'AI-Powered Healthcare Fraud Detection' in a smaller, lighter font. The main login area is a white rounded rectangle with a light purple border. It contains the heading 'Welcome Back' in bold. Below this are two input fields: 'Username' with a placeholder 'Enter your username' and 'Password' with a placeholder 'Enter your password'. A prominent purple 'Sign In' button is positioned below the password field. At the bottom of the login area, there is a 'Demo Credentials' section with a light blue background, showing 'Username: admin' and 'Password: fraud2025'. The footer at the very bottom of the screen reads '© 2025 FraudGuard. All rights reserved.'

Dashboard



FraudGuard

AI-Powered Detection

Dashboard

Provider Directory

Manual Analysis

System Architecture

QUICK LOOKUP

Enter NPI...

Dashboard Overview

Providers Monitored

1,767,599

High Risk Alerts

10,418

Avg. Fraud Score

0.232

Active Investigations

0

Priority Watchlist

View All

PROVIDER NPI	SPECIALTY	RISK STATUS	ACTION
 1053387704 ID #7704	Allergy/ Immunology	90% Risk 0.90	Analyze
 1285933754 ID #3754	Pharmacist	82% Risk 0.82	Analyze
 1912506270 ID #6270	Pharmacist	82% Risk 0.82	Analyze
 1124759980 ID #9980	Pharmacist	82% Risk 0.82	Analyze
 1114436201 ID #6201	Pharmacist	82% Risk 0.82	Analyze
 1710066709 ID #6709	Pharmacist	82% Risk 0.82	Analyze
 1639508319 ID #0319	Pharmacist	82% Risk 0.82	Analyze
 1477200111 ID #0111	Pharmacist	82% Risk 0.82	Analyze

Provider Directory



FraudGuard

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QUICK LOOKUP

Enter NPI...

Provider Directory

admin

A

Logout

Search providers (Name, NPI, or Description)...

Semantic Search

NPI	PROVIDER NAME	SPECIALTY	COST/SERVICE	RISK STATUS	ACTION
Filter NPI...	Filter Name...	Filter Specialty...			
1003000126	Ardalan Enkeshafi	Hospitalist	\$19.50	Low (0.08)	Analyze
1003000134	O O	Pathology	\$0.45	Low (0.03)	Analyze
1003000142	Rashid Khalil	Anesthesiology	\$22.84	Medium (0.38)	Analyze
1003000167	Julio Escobar	Dentist	\$3.87	Low (0.17)	Analyze
1003000423	Jennifer Velotta	Obstetrics & Gynecology	\$81.22	Low (0.27)	Analyze
1003000480	Kevin Rothchild	General Surgery	\$21.86	Low (0.08)	Analyze
1003000530	Amanda Semonche	Internal Medicine	\$55.18	Medium (0.40)	Analyze
1003000597	Dae Kim	Urology	\$71.16	Low (0.23)	Analyze
1003000639	Peyman Benharash	Cardiac Surgery	\$101.41	Medium (0.51)	Analyze
1003000704	O O	Anesthesiology	\$23.23	Low (0.07)	Analyze



FraudGuard

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QUICK LOOKUP

Enter NPI...

Investigation Report

2.2 Key Risk Drivers

The risk score was influenced by several factors. The following components had the most significant impact:

- cost_per_service:** (+0.408) This was the primary driver, contributing substantially to the high-risk score due to extreme deviation from peer averages.
- spb_z_tanh:** (+0.136) This factor, representing a statistical measure of services per beneficiary relative to peers, was a secondary contributor to the elevated risk.
- has_part_b:** (-0.019) The provider's Part B participation served as a minor mitigating factor but did not meaningfully offset the primary risk indicators.

3.0 Conclusion

The investigation confirms that Provider 1477200111 exhibits billing practices that are highly anomalous and indicative of potential fraud, waste, or abuse. The combination of an automated alert for excessive cost per service and a subsequent high-risk score of 0.8151 provides a strong basis for immediate intervention.

4.0 Recommendation

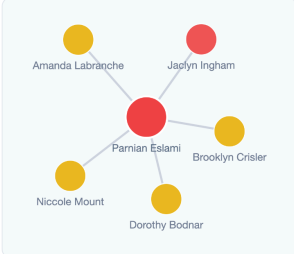
Based on the high degree of confidence in the risk assessment, the following actions are recommended:
HOLD PAYMENT: Immediately place a hold on all pending and future payments to Provider 1477200111.
INITIATE MANUAL REVIEW: Assign this case for a comprehensive manual review and audit of the provider's claims history to verify billing compliance and medical necessity.

SHAP values...

Analyst determined risk level: High (Score: 0.8151)

Report: Reporter generating final narrative...

PEER SIMILARITY NETWORK



Real-Time Analysis: Showing providers with similar billing patterns (Cosine Similarity).



FraudGuard

AI-Powered Detection

Dashboard

Provider Directory

Manual Analysis

System Architecture

QUICK LOOKUP

Enter NPI...

Investigation Report

Risk Analysis (AnalystAgent)

Risk Score Analysis

Overall Assessment: **HIGH RISK** - Immediate investigation recommended

Risk Score: **0.8151**

Key Risk Drivers

The following features contributed most significantly to this risk assessment:

- cost_per_service:** **+0.408** impact on risk score
- spb_z_tanh:** **+0.136** impact on risk score
- has_part_b:** **-0.019** impact on risk score

Network & Anomalies (InvestigatorAgent)

Anomaly Detection Results

Summary: **1 anomaly detected**

Provider: Parnian Eslami - Pharmacist

Cost/Service: \$6043.91 | Services/Beneficiary: 3.27

Detailed Findings

Unusually high cost per service exceeding \$200 threshold

FEEDBACK FOR TRAINING

RL Reinforcement

Help improve the model by verifying this analysis.

Confirm Fraud

False Positive

Suspicious

Dismiss

Submit to Supervisor

Appendix

Mathematical Notation Guide

This section explains all mathematical formulas, symbols, and statistical concepts used throughout the documentation.

Common Symbols

Symbol	Meaning	Example
μ (mu)	Mean (average) value	$\mu = \Sigma x / n$
σ (sigma)	Standard deviation	$\sigma = \sqrt{(\Sigma (x - \mu)^2 / n)}$
Σ (capital sigma)	Sum of all values	$\Sigma x = x_1 + x_2 + \dots + x_n$
$\in$ (element of)	Value belongs to set	$x \in [0, 1]$ means x is between 0 and 1
$\cap$ (intersection)	Common elements	$A \cap B$
$\cup$ (union)	Combined elements	$A \cup B$
$\rightarrow$ (arrow)	Approaches or maps to	$x \rightarrow y$
$\geq$ (greater or equal)	Greater than or equal to	$x \geq 5$
$\leq$ (less or equal)	Less than or equal to	$x \leq 10$
∂ (partial derivative)	Rate of change	$\partial L / \partial W$
$\sqrt{}$ (square root)	Square root	$\sqrt{16} = 4$

Statistical Formulas

1. Mean (Average)

$$\mu = \Sigma x / n$$

Explanation: Sum all values (Σx) and divide by the count (n).

Example: For values [10, 20, 30], $\mu = (10+20+30)/3 = 20$

2. Standard Deviation

$$\sigma = \sqrt{(\Sigma (x - \mu)^2 / n)}$$

Explanation:

1. Calculate mean (μ)
2. For each value (x), find difference from mean: $(x - \mu)$
3. Square each difference: $(x - \mu)^2$
4. Sum all squared differences: $\Sigma (x - \mu)^2$
5. Divide by count: $/ n$

6. Take square root: $\sqrt{\quad}$

Purpose: Measures how spread out values are from the mean.

Example: For [10, 20, 30] with $\mu=20$:

- Differences: [-10, 0, +10]
- Squared: [100, 0, 100]
- Sum: 200
- Divide by 3: 66.67
- Square root: $\sigma = 8.16$

3. Z-Score (Standard Score)

$$z = (x - \mu) / \sigma$$

Explanation:

- x : The value we're measuring
- μ : Mean of all values in the group
- σ : Standard deviation of the group
- z : How many standard deviations away from mean

Purpose: Normalizes values to compare across different scales.

Interpretation:

- $z = 0$: Value is exactly at the mean
- $z = 1$: Value is 1 standard deviation above mean
- $z = -1$: Value is 1 standard deviation below mean
- $|z| > 2$: Unusual value (outlier)
- $|z| > 3$: Very unusual value (strong outlier)

Example: If a provider's cost per service is \$250:

- Specialty mean (μ) = \$100
- Specialty std dev (σ) = \$50
- $z = (250 - 100) / 50 = 3.0$
- Interpretation: This provider charges 3 standard deviations above average (very high!)

4. Tanh (Hyperbolic Tangent)

$$\tanh(x) = (e^x - e^{-x}) / (e^x + e^{-x})$$

Explanation: A mathematical function that squashes any value into the range [-1, 1].

Simplified:

- Very negative numbers $\rightarrow -1$
- Zero $\rightarrow 0$
- Very positive numbers $\rightarrow +1$

Purpose: In our system, we use $\tanh(\text{z-score})$ to:

1. Keep all features in a bounded range
2. Prevent extreme values from dominating the model
3. Make features comparable

Example:

- $\tanh(0) = 0$ (at mean)
- $\tanh(1) \approx 0.76$ (above mean)
- $\tanh(3) \approx 0.995$ (very high, but bounded)
- $\tanh(-3) \approx -0.995$ (very low, but bounded)

Machine Learning Formulas

5. Cosine Similarity

$$\text{cosine_sim}(A, B) = (A \cdot B) / (||A|| \times ||B||)$$

Explanation:

- $A \cdot B$: Dot product $= A_1 \times B_1 + A_2 \times B_2 + \dots + A_n \times B_n$
- $||A||$: Magnitude of $A = \sqrt{(A_1^2 + A_2^2 + \dots + A_n^2)}$
- $||B||$: Magnitude of $B = \sqrt{(B_1^2 + B_2^2 + \dots + B_n^2)}$

Purpose: Measures how similar two providers are based on their features.

Range: -1 to 1

1: Identical behavior 0: No similarity -1: Completely opposite

Example: Two providers with features [100, 20] and [200, 40]:

- Dot product: $100 \times 200 + 20 \times 40 = 20,800$

- Magnitude A: $\sqrt{(100^2 + 20^2)} = \sqrt{10,400} = 101.98$
- Magnitude B: $\sqrt{(200^2 + 40^2)} = \sqrt{41,600} = 203.96$
- Similarity: $20,800 / (101.98 \times 203.96) = 0.999$ (nearly identical!)

6. PageRank

$$PR(v) = (1-d)/N + d \times \sum (PR(u) / L(u))$$

Explanation:

- $PR(v)$: PageRank score for provider v
- d : Damping factor (usually 0.85) - probability of following a link
- N : Total number of providers
- $PR(u)$: PageRank of provider u who links to v
- $L(u)$: Number of outgoing links from u
- Σ : Sum over all providers u that link to v

Purpose: Identifies influential providers in the similarity network.

Interpretation:

- High PageRank: Provider is similar to many other providers (central in network)
- Low PageRank: Provider has unique behavior (isolated in network)

Example: If provider A is similar to 10 others who are each similar to 2 providers, A gets importance from being a "hub" in the network.

7. Sigmoid Function

$$\sigma(x) = 1 / (1 + e^{-x})$$

Explanation: Converts any number to a probability between 0 and 1.

Simplified:

- Very negative $x \rightarrow 0$
- $x = 0 \rightarrow 0.5$
- Very positive $x \rightarrow 1$

Purpose: Final layer of neural network converts score to fraud probability.

Example:

- $\sigma(-5) \approx 0.007$ (low fraud risk)
- $\sigma(0) = 0.5$ (boundary)

- $\sigma(5) \approx 0.993$ (high fraud risk)

8. Focal Loss

$$FL(p_t) = -\alpha \times (1 - p_t)^\gamma \times \log(p_t)$$

Explanation:

- p_t : Predicted probability for the true class
- α : Weighting factor (balances positive/negative examples)
- γ : Focusing parameter (default 2.0) - emphasizes hard examples
- $\log$: Natural logarithm

Purpose: Loss function that focuses training on hard-to-classify examples.

Why We Use It:

- Standard datasets have few fraud cases (imbalanced)
- Focal loss reduces loss for easy examples (well-classified)
- Increases loss for hard examples (misclassified)
- Forces model to learn from difficult fraud cases

Example:

- Easy example ($p_t = 0.9$): Loss is reduced by factor $(1-0.9)^2 = 0.01$
- Hard example ($p_t = 0.6$): Loss is reduced by factor $(1-0.6)^2 = 0.16$
- Model focuses 16x more on the hard example!

9. SHAP (Shapley Values)

$$\phi_i = \sum [\binom{|S|}{|S|-1} \times \binom{|N|-|S|-1}{1} / |N|!] \times [f(S \cup \{i\}) - f(S)]$$

Explanation:

- ϕ_i : SHAP value for feature i (contribution to prediction)
- S : A subset of features (a "coalition")
- N : All features
- $|S|$: Number of features in subset S
- $!$: Factorial (e.g., $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$)
- $f(S \cup \{i\})$: Model prediction with features S and feature i
- $f(S)$: Model prediction with only features S

- Σ : Sum over all possible subsets S

Purpose: Fairly distributes the prediction credit among all features.

Simplified Interpretation:

- Positive SHAP: Feature increases fraud risk
- Negative SHAP: Feature decreases fraud risk
- Magnitude: How much the feature matters

Example: If base risk is 0.1 and prediction is 0.8:

- Total increase: 0.7
- SHAP values distribute this: [+0.4 from cost, +0.2 from volume, +0.1 from centrality]
- Sum of SHAP values = 0.7 (explains the full difference)

Neural Network Formulas

10. Forward Propagation

$$z = W \times x + b$$

$$a = \text{activation}(z)$$

Explanation:

- x : Input features (11-dimensional vector)
- W : Weights matrix (learned during training)
- b : Bias term (learned during training)
- z : Pre-activation value (weighted sum)
- a : Activated output (passed through activation function)

Example for one neuron:

- Input: $x = [0.5, 0.3, 0.8]$
- Weights: $W = [0.2, -0.4, 0.6]$
- Bias: $b = 0.1$
- $z = 0.5 \times 0.2 + 0.3 \times (-0.4) + 0.8 \times 0.6 + 0.1 = 0.1 - 0.12 + 0.48 + 0.1 = 0.56$
- $a = \text{PReLU}(0.56) = 0.56$ (positive, so pass through)

11. Backpropagation (Gradient Descent)

$$\partial L / \partial W = (\partial L / \partial a) \times (\partial a / \partial z) \times (\partial z / \partial W)$$

$$W_{\text{new}} = W_{\text{old}} - \alpha \times \partial L / \partial W$$

Explanation:

- L : Loss (error) of the model
- $\partial L / \partial W$: Gradient (how much changing weight W affects loss)
- α : Learning rate (step size for updates)
- Chain rule: Calculate gradient by multiplying derivatives

Purpose: Updates weights to reduce prediction errors.

Simplified Process:

1. Make prediction (forward pass)
2. Calculate error (loss)
3. Find how each weight contributed to error (backward pass)
4. Adjust weights to reduce error (gradient descent)

12. Adam Optimizer

$$\begin{aligned} m_t &= \beta_1 \times m_{t-1} + (1 - \beta_1) \times g_t \\ v_t &= \beta_2 \times v_{t-1} + (1 - \beta_2) \times g_t^2 \\ W &= W - \alpha \times m_t / \sqrt{v_t + \epsilon} \end{aligned}$$

Explanation:

- g_t : Gradient at timestep t
- m_t : Moving average of gradients (momentum)
- v_t : Moving average of squared gradients (variance)
- β_1 : Momentum decay rate (typically 0.9)
- β_2 : Variance decay rate (typically 0.999)
- α : Learning rate (typically 0.001)
- ϵ : Small constant to prevent division by zero (typically 10^{-8})

Purpose: Adaptive learning rate optimizer that: (Combines benefits of momentum and adaptive learning rates.

1. Maintains momentum (smooths out updates)
2. Adapts learning rate per parameter
3. Converges faster than standard gradient descent